Data Science for Transportation (2025) 7:28
https://doi.org/10.1007/s42421-025-00141-5

RESEARCH

A Fully Neural Network-Based Travel Demand and Scheduling Model, Covering Activities, Destinations, and Modes of Transportation

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Received: 11 July 2025 / Revised: 4 November 2025 / Accepted: 5 November 2025 / Published online: 2 December 2025
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Abstract
Recent advancements in machine learning offer promising alternatives to traditional theory-driven approaches in activity-based modeling of travel demand and scheduling. While some scheduling models have integrated combinations of activity, destination, or mode choice, none, to the best of our knowledge, have integrated all three components within a unified, fully neural network-based framework. To address this gap, this paper presents Skyline-NN, a novel, fully neural network-based model designed to simulate full-day travel and activity schedules, encompassing activities, destinations, and transportation modes. At each time step, the model system operates through three sequential sub-models, Trip Generation, Trip Distribution, and Mode Choice, using a utility-maximizing microsimulation approach. A key component is the scalable Zone Block Module, which enables evaluation of a vast number of destinations through shared parameters. This is demonstrated in the Stockholm case study, where the model handles 1375 available destinations. The model is trained on Stockholm survey data and evaluated through simulated daily schedules against a held-out test set. Its performance is benchmarked against traditional multinomial logit (MNL) baselines, showing clear improvements in predictive accuracy. The analysis further examines sequential error propagation to assess the stability of downstream predictions, includes an ablation study on mandatory destination travel time and cost features, and tests transferability using data from Helsinki. Results demonstrate the model's effectiveness in simulating travel demand and activity scheduling, laying a foundation for fully neural network-based approaches that integrate timing, activity purposes, destinations, and modes of transportation.

Keywords Neural networks · Travel demand modeling · Scheduling · Transportation planning · Simulation

Introduction

Travel demand modeling plays a central role in transportation planning and policy-making. These models are essential tools for predicting travel patterns, evaluating infrastructure projects, and assessing policy interventions such as congestion pricing or public transport incentives (Castiglione et al. 2015). Traditionally, the field has relied heavily on theory-driven approaches, notably the multinomial logit model (MNL) and its many extensions, such as the nested logit model, to understand and predict travel behavior (McFadden 1974, 1978).

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However, due to the complexity of travel behavior, traditional models with linear utility functions often struggle to accurately capture the true data-generating process. Recent advances in neural networks present a promising alternative, offering improved predictive capabilities through their data-driven ability to model nonlinear relationships (Goodfellow et al. 2016).

Despite these advancements, neural network-based travel demand models have primarily focused on single components of travel behavior, such as activity choice, destination choice, or transportation mode choice (Rasouli and Nikraz 2013; Salas et al. 2022), with fewer also incorporating scheduling (Koushik et al. 2023; Zhang et al. 2024). While a few exceptions integrate two components (Bei et al. 2023), to the best of our knowledge, no existing models have integrated all three components into a fully neural network-based scheduling framework. A natural limitation of this approach is that it fails to capture the full spectrum of an individual's activity-travel pattern throughout the day.

To address this gap, this paper proposes Skyline-NN, a fully neural network-based scheduling and travel demand model that models an agent's daily trajectory across activities performed, destinations visited, and transportation modes used. By modeling the entire activity-travel pattern, Skyline-NN accounts for both the temporal and spatial challenges inherent in such decisions.

Following the sequential approach of a four-step model (FSM) McNally (2007), the model system operates through three sub-models: Trip Generation, Trip Distribution, and Mode Choice in sequential order. Examples of models using similar sequential decision-making processes over activities, destinations, and mode choice include ALBATROSS, developed by Arentze et al. (2000), and the Data-Driven Activity Scheduler (DDAS), developed by Drchal et al. (2019). However, Skyline-NN advances this approach by integrating a fully neural network-based framework, as well as other methodological differences. These differences are more thoroughly discussed in the literature review in Sect. 2.

Besides its sequential approach, Skyline-NN deviates from traditional FSMs in its broader aspects, employing a fully utility-maximizing microsimulation approach with a detailed temporal window and nuanced neural network-based decision-making. More specifically, the model is trained to capture individuals' activity-travel decisions at any discrete time step throughout the day. In any given time step, the Trip Generation sub-model first decides whether an agent stays and continues their current activity in the same place, or initiates an activity-defined trip by selecting any of the optional activity purposes. If a trip is selected, the destination and mode choice are sequentially decided from the Trip Distribution and Mode Choice sub-models, respectively. However, the Trip Distribution sub-model only models free-time activities and is bypassed for mandatory activities, such as *home* and *work*, since the locations for these are fixed and known for each agent.

A key feature contributing to the performance and scalability of Skyline-NN is the design of the Trip Distribution sub-model. Rather than letting all input features from all zones connect to the same hidden layer, each zone is evaluated separately in its own Zone Block Module before the utilities of all zone evaluations enter the multinomial logit function, also known as softmax. However, all blocks have an identical design and utilize a shared parameter set, greatly enhancing generalization. The proposed parameter-sharing design, conceptually similar to the parameter sharing used in convolutional neural networks, allows the model to efficiently scale to large spatial systems and remain responsive to changes in accessibility or land use.

Furthermore, incorporating features that provide insights on upcoming mandatory travel times and costs to the home or work destination from visiting a zone further contributes to the accuracy of the Trip Distribution sub-model. This approach aligns with the findings from Kitamura and Kermanshah (1984), which demonstrated that the location of an individual's home and the spatial constraints imposed by the home location significantly influence non-home-based destination choices. An ablation experiment was conducted to isolate the impact of these mandatory travel time and cost features, comparing cross-entropy loss between models with and without these inputs. For more detail, see Sect. 3.5.1.

The model was trained on Stockholm survey data and further evaluated by simulating agents in a test set, comparing their daily schedules against observed ground truth counterparts. To benchmark the model against conventional approaches, corresponding multinomial logit (MNL) models were also estimated for each sub-component, showing that the neural network achieved consistently better predictive performance across all tasks. Sequential error propagation was further quantified to examine how upstream prediction uncertainty affects downstream outcomes. In addition, the model was applied to a dataset from Helsinki to demonstrate its usability and transferability.

A preliminary study on the Trip Distribution sub-model, presented by Fredriksson and Karlström (2023), evaluated its loss metric performance under different levels of non-linearity in its feature connections. An early version of the Skyline-NN model system was briefly mentioned for context but was not analyzed in detail. This paper significantly extends that work by thoroughly describing the complete Skyline-NN model, detailing the training and evaluation procedures, benchmarking the model against corresponding MNL baselines, quantifying sequential error propagation, presenting results from simulated daily schedules of agents, evaluating the contribution of forward-looking mandatory travel time and cost features through an ablation test, and performing a transferability analysis using Helsinki data.

Literature Review

Since Chapin (1974) theorized that travel demand is derived from activity participation, the travel demand modeling field shifted from trip-based, via tour-based, to activity-based (Timmermans 2005). Activity-Based Models (ABMs) focus on individuals' needs and desires to engage in activities and consider behavioral theories on decision-making processes about activity participation under various constraints. These constraints include the timing, location, and mode of transportation for the activities (Castiglione et al. 2015; Hägerstrand 1970). ABMs assume that travel causes disutility and is only used when the net utility of traveling and activity participation combined is greater than performing an activity without travel (Bowman and Ben-Akiva 2001). These models are also characterized using a disaggregated micro-agent

approach, modeling trips at an individual or sometimes household level. Activities are planned sequentially over a more detailed temporal window compared to trip-based modeling (Castiglione et al. 2015; Chu et al. 2012).

Within activity-based travel demand modeling, discrete choice models (DCMs) grounded in random utility maximization (RUM) have emerged as the predominant methodology. The multinomial logit model (MNL) and its extensions of different nested logit models are some of the most frequently employed modeling approaches within this framework (McFadden 1974, 1978). While the MNL function can be used in different modeling approaches, for example, neural networks for classification (Goodfellow et al. 2016), its terminology within the choice modeling field conventionally refers to the theory-driven modeling approach.

The conventional MNL is characterized by its linear utility function, where each variable is explicitly defined, making the interpretation of each parameter convenient. Although these models can capture some degree of nonlinearity through their exponential transformations of utilities into choice probabilities, researchers may find it useful to allow parameters to enter the representative utility nonlinearly. A consequence of this is that estimation then becomes more difficult, as the log-likelihood function may not be globally concave. Computer routines for such models are less widely available than for logit models with linear-in-parameters utility (Train 2009). Another approach to capture nonlinear relationships in the model is to include nonlinear variable specifications, such as $x_1 * x_2$ or ${x_1}^2$, while still keeping the parameters linear (Bentz and Merunka 2000). However, the numerous combinations of optional transformation functions can make suitable selections difficult. A subjective specification approach could introduce bias, resulting in the model being a poor descriptor of the actual underlying data-generating process (Cranenburgh et al. 2022).

Neural networks can address the limitations of the more theory-driven nonlinear model specifications and offer some advantages when applied to choice models: (1) While a classification-specified neural network without hidden layers closely resembles a conventional MNL model (Bentz and Merunka 2000), including hidden layers between input features and output class utilities allows for complex utility functions, with nonlinear transformations within and between features. (2) Neural networks are data-driven, enabling them to directly learn the relationships between features from the data with minimal prior assumptions. This also makes them flexible and easily modified to accommodate any additional input features or new data sources. (3) Through their hidden layers, neural networks can capture the most meaningful representations in the data. By configuring these layers, the network can reduce overfitting and enhance the model's ability to generalize (Goodfellow et al. 2016).

However, a consequence of using hidden layers is that it makes the interpretation of parameters less clear. The functions are typically complex and often referred to as a "black box" lacking transparency as to how they reach different conclusions.

Several techniques have been proposed to address the issue of explainability, which is crucial for policy analysis. Techniques such as feature importance (Molnar et al. 2021), partial dependence plots (Zhang et al. 2018), and Layer-wise Relevance Propagation (LRP) Alwosheel et al. (2021) provide ways to assess and visualize the impact of individual input features on a model's predictions. To maintain some of the interpretability associated with the conventional MNL, Sifringer et al. (Sifringer et al. 2018) proposed a hybrid model. One component represents a linear utility DCM for interpretable features, while all other features are processed through a nonlinear neural network. The neural network's output is then added as a single term to the DCM's utility function.

Another concern with the nonlinear data-driven approach is its ability to accurately predict outcomes for input feature ranges outside the support of the training data. This can result in unreliable predictions that may not be suitable for informing policy decisions (Aboutaleb et al. 2021). For scenario analysis, it is, therefore, a good practice to verify that the model responds smoothly and plausibly to changes in key input features around the scenario of interest, for example through sensitivity analysis.

While the aforementioned challenges present considerations, neural network travel demand models offer significant potential given the numerous benefits previously mentioned. Their ability to closely align with the true underlying data-generating process frequently leads to enhanced predictive performance over traditional theory-driven methods (Zhao et al. 2020; Pineda-Jaramillo 2019; Hagenauer and Helbich 2017; Omrani 2015).

Several activity scheduling models, including destination and mode choice, have been proposed over the years. Bowman and Ben-Akiva (2001) developed a discrete choice model system using nested logit. An individual's daily activities and associated travel are structured as an activity pattern composed of a set of tours. Tour models consider the choice of time of day, destination, and mode of travel and are conditioned by the choice of activity pattern.

The dynamic discrete choice model Scaper, developed by Västberg et al. (2019), constructs activity schedules based on sequential decisions of when, where, why, and how to travel. The model employs a Markov decision process framework to base decisions at each time step on their expected future utilities, building upon the work of Karlström (2005). This forward-looking approach enables the model to effectively handle the time-space constraints present in an individual's upcoming activity-travel planning.

More closely related to the sequential decision-making process of Skyline-NN is ALBATROSS (Arentze et al. 2000), an activity-based model that uses data-driven decision trees to sequentially schedule fixed and flexible activities, along with their timing, location, and mode of transportation.

While a joint approach can be advantageous in capturing complex interdependencies among all input features and alternatives, a step-by-step structure can be both computationally more efficient and easier to interpret. Furthermore, in scenarios where data is sparse, a sequential approach can reduce the dimensionality of the problem by distributing input features across different stages of the model. This approach mitigates the risk of overfitting by allowing each model to focus on a smaller, more relevant subset of features, thereby enhancing the model's ability to generalize, as supported by the principles outlined in James (2013).

A more recent sequential activity scheduler that incorporates both location and mode choice is the Data-Driven Activity Scheduler (DDAS) Drchal et al. (2019). DDAS operates through four sequential sub-models: (1) A decision tree model first decides the activity type. (2) A regression tree then sets the duration of the selected activity. (3) An activity attractor model, utilizing a neural network, selects the destination. (4) Finally, a mode choice model, again using a decision tree, selects the mode of transportation.

A key difference between DDAS and Skyline-NN lies in their scheduling processes. DDAS uses a separate sub-model to decide the activity duration after the activity type has been chosen. In contrast, Skyline-NN operates in discrete time steps, where the time of day input feature informs the Trip Generation sub-model to either initiate a new activity-based trip or choose a *stay* alternative to extend the current activity for another time step, thereby integrating duration into the scheduling process.

While both models utilize neural networks and apply a shared parameter set for destination selection, their implementations and training methodologies differ significantly. Skyline-NN employs a straightforward supervised learning approach, training the network with labeled data to predict visited zones based on features such as travel times, costs, and land-use data. In contrast, DDAS uses a multi-objective optimization approach, training its network on a subset of locations to minimize differences in distance and visit frequency between predictions and observed data.

In addition, Skyline-NN also implements future mandatory travel considerations for a zone, and uses land-use data (total population, employment retail, etc.) as attractor variables rather than visit frequency. Furthermore, DDAS selects a set of location candidates in case a location would have no feasible connection for the chosen mode of transport. In contrast, Skyline-NN directly integrates mode-specific travel times and costs into both its Trip Destination and Mode Choice sub-models, ensuring selections of viable transportation options for each destination.

The Model

This section provides an overview of the proposed Skyline-NN model. Initially, the illustrations and transformation functions used for each sub-model are described, followed by how the model system is configured. Finally, each sub-model is described separately, along with a discussion on the model system's feature selection and sequential aspects.

The input features and output alternatives shown in the illustrations and sub-model descriptions reflect the model implementation of the Stockholm dataset in Sect. 4, but these can be easily modified, reduced, or extended to suit different applications or data requirements.

Model Illustrations

The Trip Generation, Trip Distribution, and Mode Choice sub-models are visually represented in Figs. 1, 3, and 4, respectively. All input features are represented as green nodes on the left side of their respective model illustrations. The red nodes in the middle layers of each model illustration correspond to the hidden nodes, while the yellow nodes on the right side represent the different alternatives in the output layer. The parameters in each illustration are separated into weights $\mathbf{W}$ and biases $\mathbf{b}$. Every pair of nodes in adjacent layers is connected with weights in each sub-model. While these parameters are not explicitly illustrated, they are described using matrix/vector notation. Their superscripts G, D, and M reference the Trip Generation, Trip Distribution, and Mode Choice sub-models, respectively. Note that the Zone Block Module is a structure shared among all zones, resulting in each of these instances using the same parameters.

Continuous input features are normalized to lie on a scale between 0 and 1 and represented in the figures as $[0, 1]$. Categorical features are represented by their set of possible values, here being $\{0, 1\}$. The activation functions used in each layer are indicated by either the sigma symbol σ for sigmoid/logistic transformations or the term *linear* for transformations that maintain their values unchanged. The output layer of every sub-model is subsequently transformed with the MNL/softmax function.

Activation and MNL/Softmax Functions

Activation functions are applied to achieve nonlinear transformations. The hidden nodes in each sub-model utilize the sigmoid activation function in Eq. 1. The output nodes instead use a linear activation function, resulting in a weighted sum of all nodes in the previous layer without applying any change to them. Each output layer is subsequently fed to the MNL/softmax function, shown in Eq. 2, to transform their values into probabilities.

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{1}$$

$$\pi_i(x) = \frac{e^{x_i}}{\sum_{j=1}^{n} e^{x_j}} \tag{2}$$

Model Configuration

While each sub-model is trained separately, all sub-models are sequentially involved in determining an agent's decision during simulation. The Trip Generation sub-model first determines whether an activity-defined trip should be taken by sampling an activity or if the agent should stay in the same zone and do the same activity for another time step by sampling the *stay* alternative. Whenever a trip is declared by sampling a non-home or non-work activity, the Trip Distribution sub-model determines its destination. Destinations for *home* and *work* activities are fixed and known for each agent. If *home* or *work* is sampled in the Trip Generation sub-model, the simulated agent travels directly to their corresponding home or work zone, bypassing the Trip Distribution sub-model. Once a trip's destination is known, the Mode Choice sub-model uses cost and travel times to the zone to predict the most likely mode of transportation. Flowcharts of the trajectory's simulation procedure and the sampling procedure at each time step are shown in Fig. 5(a) and 5(b), respectively. Descriptions of the features included in each sub-model, with each feature's vector size and possible values, are shown in Table 1.

Table 1 This table presents the specific features, vector sizes (Size), and possible values for each sub-model: Trip Generation (Trip gen), Trip Distribution (Trip dist), and Mode Choice (Mode), as used in the sub-model illustrations and the Stockholm model implementation

Feature	Values	Size	Trip gen	Trip dist	Mode
Travel time (origin zone)	[0, 1]	3		✓	
Travel time (origin mandatory)	[0, 1]	3		✓	
Travel time (zone mandatory)	[0, 1]	3		✓	
Car cost [a]	[0, 1]	3		✓	
Land use data	[0, 1]	5		✓	
Activity to be performed	[0, 1]	4		✓	
Car access	{0, 1}	1		✓	✓
PT pass	{0, 1}	1		✓	✓
Left home w/ car	{0, 1}	1		✓	✓
Travel time destination	[0, 1]	3			✓
Car cost destination	[0, 1]	1			✓
Current activity	{0, 1}	6	✓		
Time of day	[0, 1]	1	✓		
Occupation	{0, 1}	1	✓		
Activity duration	[0,1]	1	✓		
Been at work	{0, 1}	1	✓		

[a] Comprises origin-zone, origin-mandatory, and zone-mandatory car costs

Model Formulation

The probability that an agent in time step t takes a trip with any non-home, non-work activity purpose, to one of $|J|$ destinations, using a specific mode, would be expressed as:

$$\begin{aligned} P(f_{t+1}, j_{t+1}, m_{t+1}) = & \underbrace{P(f_t \mid t, a_t, w_t, o_t, c_t)}_{\text{Trip Generation}} \cdot \\ & \underbrace{P(j_t \mid f_t, z_t, h_t, e_t, \mathbf{L}_t, \mathbf{R}_t, \mathbf{Y}_t, \mathbf{Q}_t, \mathbf{G}_t)}_{\text{Trip Distribution}} \cdot \\ & \underbrace{P(m_t \mid z_t, h_t, e_t, n_t, \mathbf{S}_t)}_{\text{Mode Choice}} \end{aligned} \tag{3}$$

where

$t \in \{1, \ldots, T\}$ is a time step.
$a \in A$ represents the current activity in the full set of activities.
$f \in A \setminus \{\text{home}, \text{work}\}$ is an activity in the set of non-home/non-work activities.
$b \in \{\text{home}, \text{work}\}$ is a home or work activity.
$j \in \{1, \ldots, J\}$ is a destination index.
$m \in M$ is a mode in the set transportation modes.
$w \in \{0, 1\}$ represents whether the agent has been to work or not.
$o \in \{0, 1\}$ represents the occupation of the individual.
$c \in [0, 1]$ represents the duration of the current activity.
$z \in \{0, 1\}$ represents whether the agent left home with car or not.
$h \in \{0, 1\}$ represents whether the agent has access to a car (at home).
$e \in \{0, 1\}$ represents whether the agent has a public transport pass.
$n \in [0, 1]$ represents the car cost to the selected destination.

Each uppercase bold matrix used in the Trip Distribution part of the equation contains $|J|$ vectors of adjustable vector length depending on the number of land-use features and mode-specific travel times used for each zone. Zone i represents the agent's current location (origin zone). j_k denotes the agent's mandatory destination (either work or home), which is predetermined for each individual. For more detail, see Sect. 3.5.1.

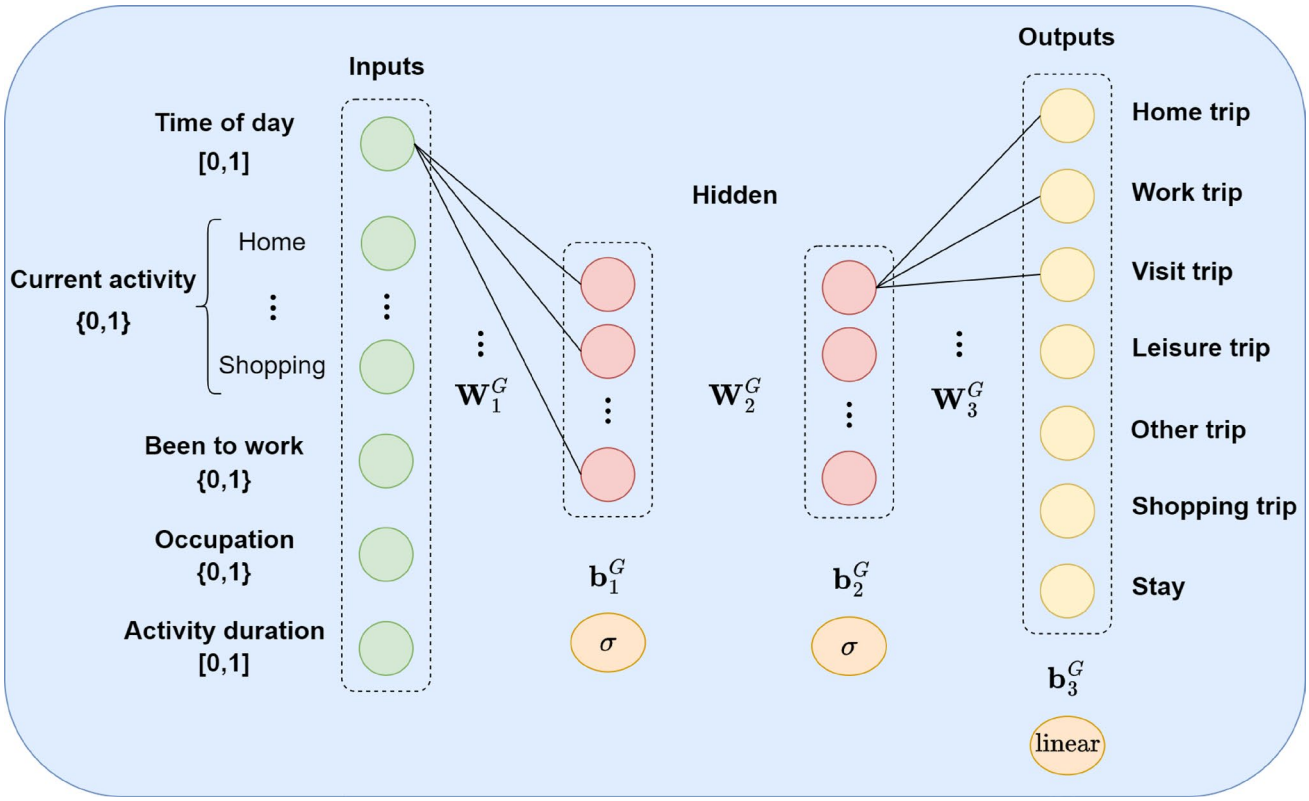


Fig. 1 Architecture of the Trip Generation sub-model

$\mathbf{L} = (l_1, \ldots, l_J)$ represents land-use data for each destination j.

$\mathbf{R} = (r_1, \ldots, r_J)$ represents travel times between destinations $i$and j.

$\mathbf{Y} = (y_1, \ldots, y_J)$ represents travel times from each destination $j$to j_k.

$\mathbf{Q} = (q_1, \ldots, q_J)$ represents travel times between $i$and j_k, identical for all j.

$\mathbf{G} = (g_1, \ldots, g_J)$ represents car cost features between $i$to j, $i$to j_k, and $j$to j_k.

$\mathbf{S} = (s_1, \ldots, s_M)$ represents mode-specific travel times from i to selected destination j_t.

Matrices $\mathbf{R}$, $\mathbf{Y}$, $\mathbf{Q}$, and $\mathbf{G}$ are dynamically updated depending on the agent's origin i and mandatory destination j_k. As seen in Eq. 3, the selected destination depends on the activity type, while the mode choice is conditional on the destination, making both the Trip Distribution and Mode Choice sub-models conditional on previous outputs.

Since the destination for any *home* or *work* activity is predetermined for each agent and exogenous to the model, the expression for any home- or work-based activity trip is reduced to:

destination is here predetermined for *home* and *work* activities, while it must be decided by the Trip Distribution sub-model for the other free-time activities.

The function mapping input to output for the Trip Generation sub-model G, here using an example of two hidden layers, is shown in Eq. 5.

The input features represented by $\mathbf{x}^{(G)}$ include *time of day*, which indicates the current time step, normalized over the T possible intervals from the start to the end of the day; *activity origin*, which represents the currently performed activity; *occupation*, which specifies whether the agent is a worker or a non-worker; *activity duration*, which measures how long the current activity has been performed; and *been to work*, which indicates whether the agent has visited work yet that day.

The output consists of either an activity-defined trip from any of the alternatives {*home*, *work*, *visit*, *leisure*, *shopping*, *other*} or *stay*. Their predicted probabilities are denoted as $\hat{\mathbf{y}}^{(G)}$ after passing through the MNL/softmax function.

$$P(b_{t+1}, j_{k,t+1}, m_{t+1}) = \underbrace{P(b_t \mid t, a_t, w_t, o_t, c_t)}_{\text{Trip Generation}} \cdot \underbrace{P(m_t \mid z_t, h_t, e_t, n_t, \mathbf{S}_t)}_{\text{Mode Choice}} \tag{4}$$

Trip Generation

The Trip Generation sub-model (Fig. 1) models whether the agent, in the next time step, chooses to stay and continue their current activity at the same location (*stay* alternative) or selects a new activity, in which case a trip will happen to a destination where that activity will be performed. The

The final architecture and training configuration for the Trip Generation sub-model, including the number of hidden layers and units, are shown in Tables 2 and 3 in Sect. 4.4. Feature values and vector sizes for each sub-model are shown in Table 1. These configurations reflect the Stockholm dataset implementation.



Fig. 2 Conceptual illustration of zone attractiveness from the agent's perspective. The figure shows how incorporating travel time to the next mandatory destination (e.g., the home zone) influences the relative attractiveness of alternative zones

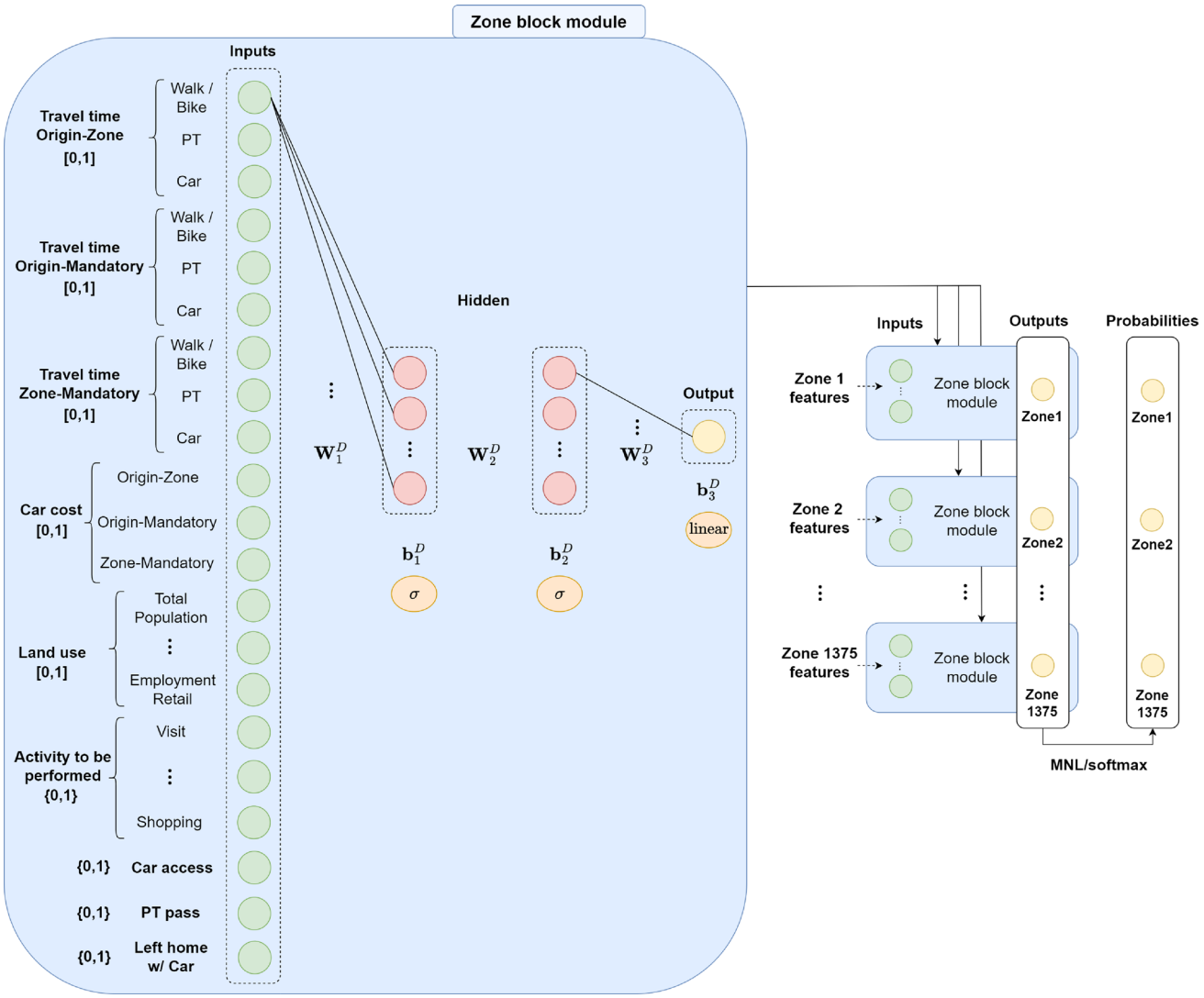


Fig. 3 Architecture of the Trip Distribution sub-model with shared Zone Block Modules

$$
\begin{aligned}
\mathbf{h}_1^{(G)} &= \sigma(\mathbf{W}_1^{(G)}\mathbf{x}^{(G)} + \mathbf{b}_1^{(G)}) \\
\mathbf{h}_2^{(G)} &= \sigma(\mathbf{W}_2^{(G)}\mathbf{h}_1^{(G)} + \mathbf{b}_2^{(G)}) \\
\hat{\mathbf{y}}^{(G)} &= \pi(\mathbf{W}_3^{(G)}\mathbf{h}_2^{(G)} + \mathbf{b}_3^{(G)})
\end{aligned} \tag{5}
$$

Trip Distribution

Whenever a trip with a free-time activity purpose is selected in the Trip Generation sub-model, the Trip Distribution sub-model determines its destination among the $|J|$ available zones, given the selected free-time activity purpose. An overview of the Trip Distribution sub-model is shown in Fig. 3. What is noticeable from the model design in the figure is that each zone $j \in J$ is evaluated in its own block, called the Zone Block Module. Consequently, the number of blocks in the sub-model equals the number of optional zones. Thus, the sub-model can, therefore, be viewed as $|J|$ of these blocks stacked on top of each other, where each block's output is further sent to the MNL/softmax function.

Each block is identical in both design and parameters, as all Zone Block Modules share the same parameter set. Given the large number of zones to be evaluated, it is computationally efficient and behaviorally reasonable to let all zones train and predict using shared parameters.

With this parameter-sharing design, the model scales efficiently along the spatial dimension and supports a large number of zones, limited only by the available computational resources. An additional benefit of this approach is that it maintains a direct link between each zone's input features and output utilities. This ensures that any changes in a zone's accessibility or land-use characteristics are directly reflected in its output alternatives, since each zone

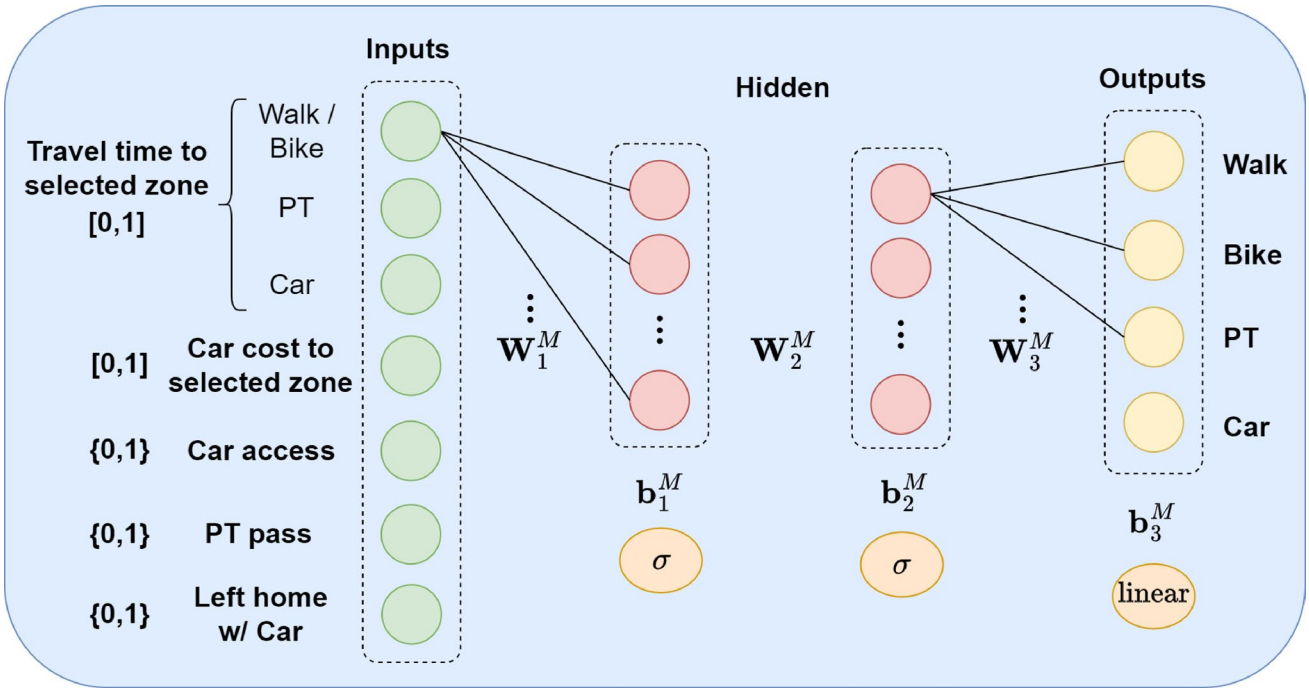


Fig. 4 Architecture of the Mode Choice sub-model

is evaluated independently through its own feature vector. In addition, the parameter-sharing design produces smooth and generalizable mappings from travel times and costs to destination utilities, making the model inherently adaptable to network interventions or land-use changes without requiring retraining for typical scenario changes.

Travel Time Sections

In each zone block, travel times input features are given in three different sections. *travel time (origin zone)* is straightforward with travel times to the evaluated zone $j \in J$, from the zone the agent is currently positioned in, the origin zone i. The other section, *travel time (zone mandatory)*, gives information on travel times from zone j to the mandatory destination j_k. This acts as a kind of forward-looking evaluation aspect, providing information on what future mandatory travel times would look like if the agent visited the zone. The mandatory destination j_k refers to the zone where the mandatory activity eventually will take place. This would be the *work* activity before a worker has arrived at work or the *home* activity in all other situations. From combining these, zones "on the way" to the mandatory zone would now be much more reasonable options.

Figure 2 illustrates this concept. The following discussion applies to the perspective of travel times but is also relevant for travel costs. Among the many zones available for potential free-time activity trips, the figure highlights a subset of these, {*Zone A*, *Zone B*}, for illustration purposes.

Currently, the agent is positioned in the origin zone i. From the perspective of the *travel time (origin zone)* feature section, both *Zone A* and *Zone B* initially appear equally attractive. However, including information from the *travel time (zone mandatory)* feature section provides additional insight, highlighting how *Zone A* has shorter travel times to the next mandatory destination, in this case, the home zone, and would from this perspective alone be a more compelling option compared to *Zone B*.

Figure 14 in Appendix A illustrates how the model effectively utilizes the mandatory travel time sections to

Table 2 Best hyperparameter results

Sub-model	LR	BS	H1	H2	VL	TL	VA	TA	E	Sec
Trip Generation	0.01	2048	32	32	0.120	0.119	0.974	0.974	238	434
Trip Distribution	0.01	8	32	32	5.183	5.029	0.139	0.138	37	313
Mode Choice	0.01	256	8	8	0.626	0.652	0.796	0.789	319	87

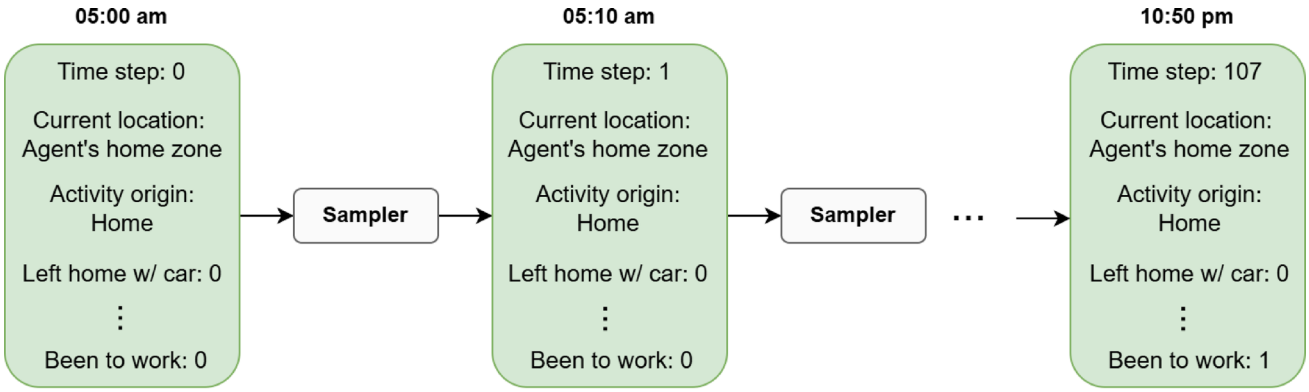
Learning rate (LR), batch size (BS), hidden layers (H1-H2), validation loss (VL), training loss (TL), validation accuracy (VA), training accuracy (TA), epoch (E), and total training time in seconds (Sec)

Table 3 Dimensions of parameters and vectors for all sub-models

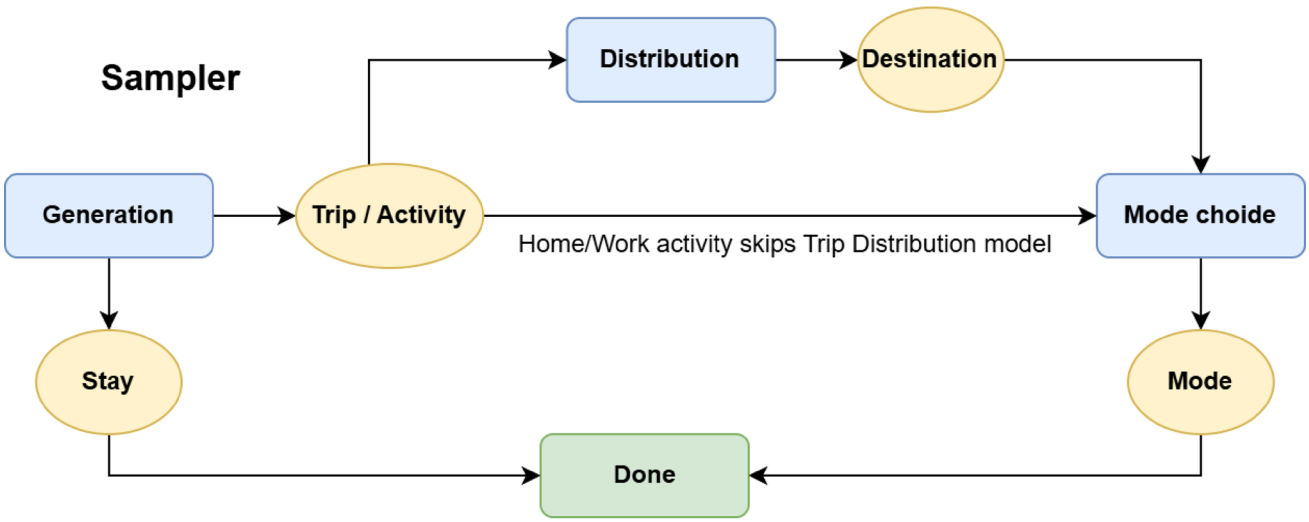
Sub-model	Input	Hidden layer 1			Hidden layer 2		
	x	W	b	h	W	b	h
Trip generation	10×1	32×10	32×1	32×1	32×32	32×1	32×1
Trip distribution	24×1	32×24	32×1	32×1	32×32	32×1	32×1
Mode choice	7×1	8×7	8×1	8×1	8×8	8×1	8×1

	Output layer		
	W	b	$\hat{y}$
Trip generation	7×32	7×1	7×1
Trip distribution	1×32	1×1	1×1
Mode choice	4×8	4×1	4×1

The specifications for the Trip Distribution sub-model refer to the operations in the Zone Block Module, with an input vector representing the features for the zone that the zone block is evaluating



(a) State space evolution process



(b) Sampling process at each time step

Fig. 5 **a** A flowchart illustrating how the state space can evolve for each time step of an agent's daily scheduling. **b** A flowchart of the sampling process for an agent at time step t choosing whether to stay or take a trip. Whenever a trip is chosen from sampling an activity, a destination is sampled for any non-home, non-work activity. The mode of transportation is finally sampled by the Mode Choice model. Adapted from (Fredriksson and Karlström 2023)

capture the underlying decision-making process. Zones close to the agent's origin, as well as zones on the way to and near the next mandatory activity, are more likely to be visited.

The third and last travel time section, *travel time (origin mandatory)*, provides information on the time it would take to travel directly from the agent's current location i to the next mandatory destination j_k for each transportation mode. This section enables the model to account for the proximity of the agent to j_k and assess a zone's features in relation to this proximity.

The contribution of these mandatory travel time sections, along with the additional mandatory cost features described in Sect. 3.5.2, was evaluated in an ablation experiment presented in the Results section.

Additional Features

Similarly to the travel time sections, car cost was given as three separate input variables: *car cost (origin zone)*, *car cost (origin mandatory)*, and *car cost (zone mandatory)*, to enable information on future mandatory travel cost. The input feature *activity to be performed* provided the free-time activity that was to be performed at the destination. *Land use* provided beneficial zone-specific descriptions when determining if the zone is attractive for the selected activity. The binary feature *left home w/ car* was activated each time the transportation mode car had been used when leaving home and if the individual or agent had not yet returned back home. *Car access* represents whether the agent has a car available at home, and the feature was only activated when the agent was home. *PT pass* indicates whether the agent holds a public transport pass, which allows for free travel on public transport during the pass's validity period. Feature values and vector sizes for each sub-model, as used in the Stockholm implementation, are shown in Table 1.

Model Function

The function mapping input to output for the Trip Distribution sub-model D, here using an example of two hidden layers, is shown in Eq. 6. The index j here corresponds to one of the $|J|$ zones being evaluated, with its respective architecture block and class alternative v. The input features to each Zone Block Module, denoted as $\mathbf{x}_j^{(D)}$, are specific for each zone j, although some of these features would be equal for all zones, such as *car access* and *PT pass*. The output vector $\hat{\mathbf{y}}^{(D)} \in \mathbb{R}^J$ contains the probabilities for each zone. All hidden and output nodes in the zone architecture block have unique biases, which are shared among all zones due to the shared Zone Block Module structure, where each instance uses the same parameters.

The final architecture and training configuration for the Trip Distribution sub-model, including the number of hidden layers and units, are shown in Tables 2 and 3 in Sect. 4.4. These configurations reflect the Stockholm dataset implementation.

$$\begin{aligned} \mathbf{h}_{1j}^{(D)} &= \sigma(\mathbf{W}_1^{(D)}\mathbf{x}_j^{(D)} + \mathbf{b}_1^{(D)}) \\ \mathbf{h}_{2j}^{(D)} &= \sigma(\mathbf{W}_2^{(D)}\mathbf{h}_{1j}^{(D)} + \mathbf{b}_2^{(D)}) \\ v^j &= \mathbf{W}_3^{(D)}\mathbf{h}_{2j}^{(D)} + \mathbf{b}_3^{(D)} \end{aligned} \tag{6}$$

The final output probabilities for all classes/zones $\mathbf{v}$ are thereafter given from the MNL/softmax activation function

$$\hat{\mathbf{y}}^{(D)} = \pi(\mathbf{v})$$

where

$$\mathbf{v} = [v^1, v^2, \ldots, v^J]^T$$

Mode Choice

The Mode Choice sub-model, shown in Fig. 4, is designed to model the choice of transportation mode to the selected zone. During simulation, this destination is either the sampled zone from the Trip Distribution sub-model for free-time activities or the home or work zone when one of these activities is selected in the Trip Generation sub-model.

The function mapping input to output for the Mode Choice sub-model M, using an example of two hidden layers, is shown in Eq. 7. Its input features $\mathbf{x}^{(M)}$ include *travel times to selected zone* (travel times for each mode), *car cost to selected zone*, *left home w/ car*, *car access*, and *PT pass*. The output vector $\hat{\mathbf{y}}^{(M)}$ gives the probability for each transportation alternative.

The final architecture and training configuration for the Mode Choice sub-model, including the number of hidden layers and units, are shown in Tables 2 and 3 in Sect. 4.4. Feature values and vector sizes for each sub-model are shown in Table 1. These configurations reflect the Stockholm dataset implementation.

$$\begin{aligned} \mathbf{h}_1^{(M)} &= \sigma(\mathbf{W}_1^{(M)}\mathbf{x}^{(M)} + \mathbf{b}_1^{(M)}) \\ \mathbf{h}_2^{(M)} &= \sigma(\mathbf{W}_2^{(M)}\mathbf{h}_1^{(M)} + \mathbf{b}_2^{(M)}) \\ \hat{\mathbf{y}}^{(M)} &= \pi(\mathbf{W}_3^{(M)}\mathbf{h}_2^{(M)} + \mathbf{b}_3^{(M)}) \end{aligned} \tag{7}$$

Feature Selection

Each sub-model was given a set of input features that seemed reasonable for their task and performance. With this in mind, any of the sub-models may have performed better with additional or fewer features. For example, the Trip Distribution sub-model may have performed slightly better with the *previous activity* feature or removed from any of its current features, such as *PT pass*.

As the number of samples in the dataset is limited, adding more input variables to each sub-model poses the risk of overfitting and worsens its generalization capabilities. At the same time, too few features may limit its ability to perform adequately. For optimal performance, it would be necessary to test all the various combinations of feature sets by measuring their validation data performance. Since each of these tests further requires a search among multiple hyperparameter settings to find the best-performing setting, it is not usually feasible to try all combinations of features (Goodfellow et al. 2016).

Sequential Decision-Making

The sequential operation of the model system is present in two ways. First, at each time step, a trip is decided sequentially, beginning with the Trip Generation sub-model, which determines whether a trip should occur. If a trip is generated, the Trip Distribution sub-model selects a destination for free-time activities, and the Mode Choice sub-model lastly determines the mode of transport to the selected destination.

Second, the scheduling of the entire day is also sequential between time steps, starting in the morning at time step $t = 1$. This approach poses a risk of compromising the interdependencies present between decisions over different time steps. To mitigate this issue, both past and future information about the agent are incorporated as state variables. Historical information, such as *been to work* and *activity duration* in the Trip Generation sub-model, and *left home w/ car* in the Trip Distribution and Mode Choice sub-models, is provided as input features. In addition, to account for future awareness, the Trip Distribution sub-model's "look-ahead" mandatory travel time and cost sections implement the consequence of future mandatory travel associated with traveling to a particular zone.

Model Implementation

Data

The data used for training and evaluation were derived from a travel questionnaire conducted in 2015 on Stockholm citizens. In total, the dataset included 10,819 IDs, where each ID was at home at both the beginning and the end of the modeled time frame, namely 5:00 am and 11:00 pm.

The day was temporally aggregated into 10-min interval time steps. Each individual's daily trajectory in the dataset was converted into samples, one for each time step, where the individual either remained at their current location and continued their activity or departed on a trip. Time steps during which an individual was exclusively traveling were not used for training any sub-model.

The Trip Generation sub-model was trained using samples of both stay and trip actions. This sub-model not only determined the activity purpose of a trip but also whether the agent remained for another time step (the *stay* alternative). The Trip Distribution sub-model, on the other hand, only considered trip samples, further excluding those with *home*/*work* activity purposes, as these zones are already known for each agent. In contrast, the Mode Choice sub-model was trained on all trip samples, regardless of activity purpose, to model the choice of transportation mode, given its destination.

The dataset included weights for each individual, reflecting whether their socio-demographic group was under or overrepresented compared to the regional population. These weights were used in both training and evaluation to ensure representativity at the population level.

The 10,819 days of data were randomly split into 3 sets using a fixed random seed (42) for reproducibility: 55% for training, 20% for validation, and 25% for testing the sub-models.

Any trip that an individual took was registered in the survey data. Each trip includes information on departure and arrival times, the zone ID of the origin and destination, the mode of transport, and the activity performed at the destination. A trip consists of a combination of the following alternatives, one from each category:

Activities: *home*, *work*, *visit*, *leisure*, *other*, and *shopping*
Zones: 1375 in total
Modes: *walk*, *bike*, *public transport (PT)*, and *car*

The land-use data used for the Trip Distribution sub-model includes the following land-use features of each zone to determine its appeal for different activities.

Land use: *total population*, *total employment*, *employment retail*, *employment restaurants*, and *employment leisure*

Every non-categorical feature in the dataset was normalized to the interval [0, 1]. Binary categorical features were represented with either 1 or 0, or one-hot encoded otherwise,

where each option in the feature category is represented by its own binary node.

Travel Time

Travel time matrices for *PT* and *car* were available for every trip between any 2 zones among the 1375 zone options. Both modes had additional peak travel time matrices that were applied for each trip between 6:30 am–9:00 am and 15:30 pm–18:00 pm. Wait time matrices and in-vehicle matrices for *PT* were aggregated to represent the trip's entire duration.

Travel times for *walk* and *bike* were derived from road network distances between zones, assuming average speeds of 4 km/h and 15 km/h, respectively. Therefore, since their travel times are both scaled versions from the same distance matrix, their normalized travel times become identical. Consequently, only one input feature node was necessary to represent both modes' travel times in the Trip Distribution and Mode Choice sub-models. The travel times matrix $\mathbf{T}_{ij}^{m}(\tau)$ represents the travel times between the origin zone i and destination zone j using mode m, either peak or off-peak hours based on the time of day τ.

Monetary Cost

In Stockholm, toll fees apply to cars depending on the zone and time of day. To account for this, two toll matrices were used: one for peak hours and another for off-peak hours. In addition, a distance-based cost of 1.53 SEK per kilometer is applied, resulting in the total car cost matrix, as calculated in Eq. 8:

$$\mathbf{C}_{ij}^{\text{car}}(\tau) = \mathbf{C}_{ij}^{\text{toll}}(\tau) + \mathbf{C}_{ij}^{\text{distance}} \tag{8}$$

Here, $\mathbf{C}_{ij}^{\text{toll}}$ represents the toll cost matrix in SEK between the origin zone i and destination zone j, which is a function of the time of day τ, reflecting peak or off-peak hours. $\mathbf{C}_{ij}^{\text{distance}}$ provides a distance-based cost between zones i and j.

The Stockholm public transport system employs a flat fare system, limiting the variability in its cost feature. Since the neural network requires variation in its features to capture their effect, the absence of this variation led to the omission of the PT cost feature from the model. Instead, a binary feature indicating the availability of a long-term *PT pass* is used as a proxy for increased PT usage.

However, in scenarios where public transport costs vary, it can be easily included as an input feature similar to car cost. This is shown in Sect. 7, where the model is trained on a Helsinki dataset that includes considerable variation in public transport costs, due to its zone-based cost system.

Training the Model System

Each of the three sub-models, Trip Generation, Trip Distribution, and Mode Choice, is a neural network trained individually using the Adam optimization algorithm (Kingma and Ba 2014). Weights were initialized using Xavier initialization (Glorot and Bengio 2010), and the selected loss function was the categorical cross-entropy loss, as shown in Eq. 9:

$$\text{Loss} = -\frac{1}{N}\sum_{s=1}^{N}\sum_{j=1}^{C} y_{j,s}\log(\hat{y}_{j,s}) \tag{9}$$

Here, $\hat{y}$ represents the predicted output probabilities, and y the desired ground truth output. s indexes over all samples in the dataset $D = \{(\mathbf{x}_s, y_s)\}_{s=1}^{N}$ with N total samples, and j indexes over each class output for a sample. This loss is finally averaged over all samples.

Python was used as the programming language for data pre-processing and to create the feature and label arrays used to train each sub-model. The deep learning library Keras was used to design, train, and predict every neural network.

To correct for the under- and over-representation of different socio-demographic groups, individual sample weights were applied during training using the `sample_weight` parameter in Keras.

The Trip Generation and Mode Choice sub-models had a straightforward design with a fully connected layer between inputs and outputs. The input feature vectors for both sub-models were structured as multi-dimensional arrays with shapes (samples, features). The total size of the feature dimension vector of each sample corresponded to the sum of the sizes of all its feature attributes, as detailed in Table 1.

To implement the shared weights concept in the Trip Distribution sub-model, the simplest and computationally fastest approach was to use a one-dimensional Convolutional Neural Network (CNN) LeCun et al. (2015). Using an input feature vector where all zones' features were stacked sequentially, the filter and stride sizes in the CNN were set equal to the number of features within a zone. This setup enabled each filter to evaluate all available information for a given zone before striding to the subsequent zone in the array to perform the same convolution operation on its features. As each filter uses the same parameters for each of its operations, the zone blocks effectively share parameters.

Since the one-dimensional CNN operates along the time dimension of a tensor, treated as the second dimension in Keras, the input feature array was reshaped to (samples, zone-features, 1). Here, zone-features represents the product of the number of features per zone (24) and the number of zones (1375), i.e., 24×1375. The CNN's output was a tensor of shape (samples, number of zones, CNN filters). The output from each CNN filter convolution on a zone's features

can be seen as producing a hidden node value for that zone, where all CNN filters collectively form its hidden layer.

For all subsequent hidden layers, a Fully Connected Layer (FCL) ("Dense layer" in Keras) was wrapped with a TimeDistributed layer. This allowed the same FCL, using the same set of parameters, to be applied to each zone's hidden layer. A final TimeDistributed FCL layer with one hidden node was used to output the utility/logit for every zone. This layer was finally flattened before the softmax function.

Hyperparameter Search

The hyperparameters were tuned through an iterative procedure aimed at minimizing the validation loss, with early stopping applied if no improvement occurred for 50 consecutive epochs. For each sub-model, the initial configuration consisted of 2 hidden layers with 32 nodes each, a learning rate of 0.001, and a batch size of 128, following commonly used defaults in the deep learning literature (Kingma and Ba 2014).

The search explored batch sizes (64, 128, 256) and numbers of hidden nodes (16, 32, 64), using powers of two, along with network depths ranging from one to three layers. Learning rates were varied across the sequence (0.001, 0.005, 0.01, 0.05). For each rate, batch size and node count were systematically adjusted until the validation loss reached a local minimum, where adjacent configurations for any non–learning-rate parameter produced higher losses. The final learning rate was then selected as the value whose surrounding rates produced consistently higher losses across all other hyperparameter combinations. For example, since two-layer architectures consistently outperformed both shallower and deeper networks in each setting, they were treated as locally optimal and no further layer depths were evaluated. The same node count was consistently applied across all hidden layers.

Due to randomized weight initialization, validation results for identical configurations showed slight variation across runs. Each sub-model was, therefore, trained 15 times with the same hyperparameter settings, and the average validation loss across these runs was used for model selection.

Preliminary tests indicated that sigmoid activations performed better than ReLU, though ReLU was tested on the best sigmoid configurations without yielding improvements.

The final hyperparameter settings and their corresponding training results are reported in Table 2. The dimensions of all parameter matrices and vectors, including those for the hidden layers, inputs, and outputs, are provided in Table 3. The training and validation loss and accuracy progressions per epoch for the best-performing sub-models are shown in Fig. 15 in Appendix B.

Technical Specifications

Training and simulation were performed on a server node equipped with an NVIDIA GeForce RTX 3080 GPU (10 GB VRAM) and 20 GB of allocated system memory. The model handled all 1375 zones simultaneously during both training and simulation without approaching these memory limits, indicating moderate resource requirements. Simulation of 1,000 individuals required approximately 59.5 min on the GPU and 56.5 min on the CPU, a difference of about 5%. This demonstrates that the simulation phase can be efficiently executed on standard CPU hardware without the need for GPU support.

Evaluation

All evaluations in this section are based on the Stockholm dataset. The same simulation procedure was later applied to the Helsinki data, as described in Sect. 7.

Comparison with Multinomial Logit Baselines

To assess the contribution of the neural network architecture, each sub-model was estimated using a traditional multinomial logit (MNL) specification. To ensure comparability, the same input features were used wherever possible. However, the continuous *time of day* feature in the Trip Generation sub-model was unsuitable for the MNL formulation and was, therefore, discretized into categorical time intervals. Specifically, *time of day* was represented using seven bins covering the period between 5 am and 11 pm: 5:00–6:30 am (early morning), 6:30–9:00 am (morning peak), 9:00–12:00 pm (late morning), 12:00–3:30 pm (early afternoon), 3:30–6:00 pm (afternoon peak), 6:00–8:30 pm (early evening), and 8:30–11:00 pm (late evening). These intervals also coincide with the peak-hour periods applied in the travel time and cost matrices.

Two versions of each sub-model were estimated and evaluated: one estimated exclusively on the training data (identical to the data used for the neural network models), and another estimated on both the training and validation datasets, reflecting standard practice in conventional MNL estimation. Both versions were evaluated on the same held-out test dataset as their neural network counterparts. The results of these comparisons are presented in Sect. 6.1. All models were trained on the same NVIDIA GeForce RTX 3080 GPU to ensure computational comparability.

Simulation of Agents' Daily Trajectories

To better understand what the trained sub-models have learned and assess their performance, the entire model system was further evaluated using simulations. By simulating a daily trajectory of activities performed, locations visited, and transportation modes used for each agent in the left-out test dataset, their activity-travel patterns can be compared with their actual observed/ground truth trajectories using various analyses and statistics.

Sampling from a sub-model is done by letting the sub-model predict on an input vector based on the agent's current state. A class is then sampled from the output vector of class probabilities. Sampling was performed sequentially at each of the 108 time steps, excluding those where a trip was still in progress.

Each agent is initially positioned at their respective home zone, performing the *home* activity at the start of the day at 5:00 am. Each working agent also has a fixed work zone ID, and whenever the activity *home* or *work* is sampled in the Trip Generation sub-model, the agent travels to their respective zones.

During simulation, a sampled trip was always considered to start in the middle of the current time step. For example, for time step 1, this would be 5:05 am, for time step 2, 5:15 am, etc. Equation 10 calculates the time step (*TS*) in which the agent arrives after initiating a trip of a certain travel time (*TT*) in minutes:

$$TS_{\text{arrival}} = TS_{\text{departure}} + \left\lfloor \frac{TT - 5}{10} \right\rfloor + 1 \tag{10}$$

A flowchart visualizing the simulation procedure of a trajectory can be seen in Fig. 5a, and a flowchart of the sampling of each action per time step is shown in Fig. 5b.

The simulation procedure is written in pseudo-code in Algorithm 1. Here, the following abbreviations are used: S_t represents the state space features at time step t; `next_act`: next activity; `next_loc`: next location; `c_act`: current activity; `c_loc`: current location; `worked`: been to work; `duration`: activity duration; `left_with_car`: left home with car; `occ`: occupation; `tt`: travel time; `cc`: car cost; `oz`: origin zone; `om`: origin mandatory; `zm`: zone mandatory.

Algorithm 1 A simulated daily trajectory for a single agent

```
c_act ← home
c_loc ← Agent's home zone ID
occ ← Agent's occupation
car_access, pt_pass ← If agent has car access or a PT pass
time_step, duration, worked, left_with_car ← 0
T ← 108                                   ▷ Number of time steps used throughout the day
while time_step < T do
    gen_out ← G(S_t)                      ▷ Sample from Trip Generation sub-model G
    if gen_out == stay then               ▷ Agent continues c_act in c_loc another time step
        mode ← none
        time_step++, duration++
    else                                                              ▷ A trip is taken
        next_act ← gen_out                                ▷ The sampled trip purpose
        if next_act == home then
            next_loc ← home zone
        else if next_act == work then
            next_loc ← work zone
            worked ← 1
        else
            tt_oz, tt_om, tt_zm, cc_oz, cc_om, cc_zm ← travel times/costs
            next_loc ← D(S_t)             ▷ Sample from Trip Distribution sub-model D
        end if
        tt_sd, cc_sd ← travel times/costs to selected destination
        mode ← M(S_t)                     ▷ Sample from Mode Choice sub-model M
        if c_act == home and mode == car then
            left_with_car ← 1
        end if
        if next_act == home then
            left_with_car ← 0
        end if
        c_act ← next_act, c_loc ← next_loc, duration ← 0
        time_step ← time_step_arrival
    end if
end while
```

Table 4 Comparison with multinomial logit (MNL) baselines and neural network (NN) models

Version	Log-loss	Accuracy	Top-5	Iterations	Train time (s)
Trip generation					
MNL (T)	0.1216	0.9742	—	396	100
MNL (T+D)	0.1215	0.9742	—	439	144
NN	0.1174	0.9742	—	238	434
Trip distribution					
MNL (T)	5.563	0.0733	0.2043	36	82
MNL (T+D)	5.564	0.0743	0.2038	44	131
NN	5.042	0.1231	0.2966	37	313
Mode choice					
MNL (T)	0.6659	0.7585	—	175	1
MNL (T+D)	0.6616	0.7612	—	194	1
NN	0.6425	0.7929	—	319	87

"T" = trained on training data only; "T+D" = trained on training+validation. Top-5 accuracy is reported for Trip Distribution only

Quantifying Sequential Error Propagation

To assess how upstream prediction errors influence downstream behavioral outcomes, sequential error propagation is quantified for the chain: *Trip Generation* (TG) → *Trip Distribution* (TD) → *Mode Choice* (MC). The analysis isolates how uncertainty in the upstream TG and TD components affects the mode-share distribution outcomes in the MC sub-model. Two simulation configurations are compared:

- **Full-chain:** The standard sequential framework, in which the MC model receives the predicted destinations generated by the TD sub-model, which itself depends on TG predictions. The resulting mode-share distribution is obtained from the full-day simulations described in Sect. 6.5.
- **OBS-D run:** A counterfactual configuration where the MC model instead receives the *observed* destinations from the test data. This setup removes the influence of upstream prediction noise and represents the case where the Mode Choice model receives the same destination distribution that underlies the observed mode choices, corresponding to perfectly accurate upstream models in the aggregate sense.

The comparison isolates the extent to which upstream errors in activity and destination prediction affect the final mode-share distribution. For each configuration, the aggregate mode-share vector $\mathbf{p} = (p_{\text{walk}}, p_{\text{bike}}, p_{\text{pt}}, p_{\text{car}})$

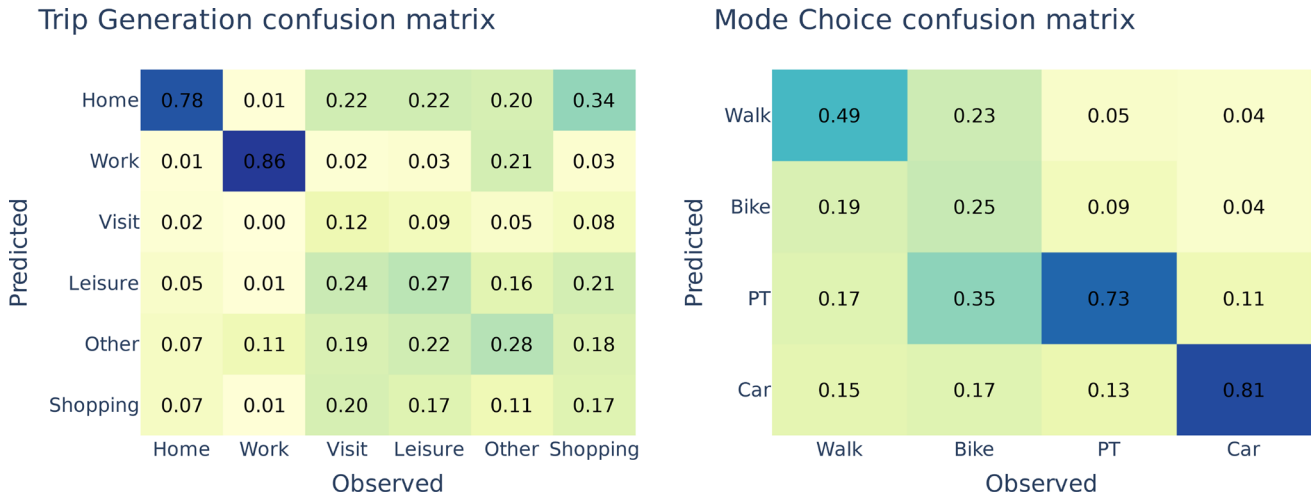


Fig. 6 Confusion matrices for the Trip Generation and Mode Choice sub-models

Table 5 Best hyperparameter results for both Trip Distribution versions; the original sub-model with all features (Original) and the version without mandatory travel time and cost features (No Mand). Learning rate (LR), batch size (BS), hidden layers (H1-H2), validation loss (VL), training loss (TL), validation accuracy (VA), training accuracy (TA), epoch (E), and total training time in seconds (Sec)

Sub-model	LR	BS	H1	H2	VL	TL	VA	TA	E	Sec
Trip Distribution (Original)	0.01	8	32	32	5.183	5.029	0.139	0.138	37	313
Trip Distribution (No Mand)	0.01	32	16	16	5.456	5.364	0.120	0.114	159	199

Table 6 Test set performance of the Trip Distribution sub-model with (Original) and without (No Mand) forward-looking mandatory travel time and cost features

Model version	Loss	Accuracy
Trip Distribution (Original)	5.042	0.123
Trip Distribution (No Mand)	5.366	0.097

Table 7 Mode share distributions under different configurations

Mode	Observed	Full-chain	OBS-D
Walk	0.123	0.126	0.120
Bike	0.103	0.095	0.103
PT	0.360	0.364	0.366
Car	0.413	0.415	0.412

TV(full-chain, observed) = 0.0085; TV(OBS-D, observed) = 0.0050; $\Delta = 41.2\%$ reduction in mode-share error

is computed and compared against the observed distribution $\mathbf{q}$ using the *total variation* (TV) distance:

$$\mathrm{TV}(\mathbf{p}, \mathbf{q}) = \frac{1}{2} \sum_m |p_m - q_m|, \tag{11}$$

which measures the total proportion of mode share that differs between the simulated and observed distributions.

The relative reduction in mode-share error when using observed destinations is given by

$$K = \frac{\mathrm{TV}_{\mathrm{full}} - \mathrm{TV}_{\mathrm{GT\text{-}D}}}{\mathrm{TV}_{\mathrm{full}}} \times 100\%, \tag{12}$$

where $\mathrm{TV}_{\mathrm{full}}$ and $\mathrm{TV}_{\mathrm{GT\text{-}D}}$ denote the total variation distances to the observed mode-share distribution for the full sequential chain and the GT-D configuration, respectively. The indicator K, thus, expresses the percentage of the total mode-share error attributable to upstream uncertainty in the predicted destinations.

Results

This section presents the results from the different evaluation approaches applied to the model system, based on the Stockholm dataset. Some analyses focus on the sub-models individually, such as the multinomial logit baselines and confusion matrices, while others assess the integrated framework, including sequential error propagation and full-day activity–travel simulations.

Comparison with Multinomial Logit Baseline Results

To benchmark the neural network architecture, each sub-model was also estimated using a corresponding multinomial logit (MNL) formulation. Table 4 presents the results, comparing the predictive performance of the neural network and MNL versions on the held-out test set, and reports the number of iterations required for convergence as well as the total training time.

The neural network architecture consistently outperforms the MNL baselines across all sub-models. The largest improvement is observed for Trip Distribution, while the gains for Trip Generation and Mode Choice are smaller but still statistically meaningful. Notably, estimating the MNL models on both the training and validation data, compared to using the training data alone, produced only small gains for mode choice, while the other sub-models showed no improvement or slightly higher losses.

It should be noted that the high accuracy observed for the Trip Generation model is largely trivial, as individuals in the dataset most frequently remain at their current location at each time step. Consequently, all model configurations predict this dominant "stay" behavior, resulting in similar accuracies at 0.9742.

While the neural network models require considerably longer training times, they achieve substantially lower losses and higher predictive accuracy. The results across all sub-models indicate that the neural network formulation is able to capture nonlinear relationships and interdependencies that the traditional MNL framework cannot represent.

Confusion Matrix Analysis

The Trip Generation and Mode Choice sub-models were further assessed using confusion matrices that illustrate the correspondence between predicted and observed classes. The Trip Distribution sub-model was omitted from this analysis, as the large number of destination alternatives (1375 zones) relative to the limited number of trips made confusion matrix visualization impractical. These tests were performed using ground truth inputs for the predictions, and the results are based on soft expected confusion, where the predicted class probabilities (rather than sampled outcomes) were aggregated across all decision points. This approach reduces variance from random sampling and highlights systematic differences between predicted and observed choices. Each matrix is column-normalized, showing the distribution of model predictions for each observed class (see Fig. 6).

The Trip Generation confusion matrix shows that the two mandatory activities, *home* and *work*, exhibit the highest

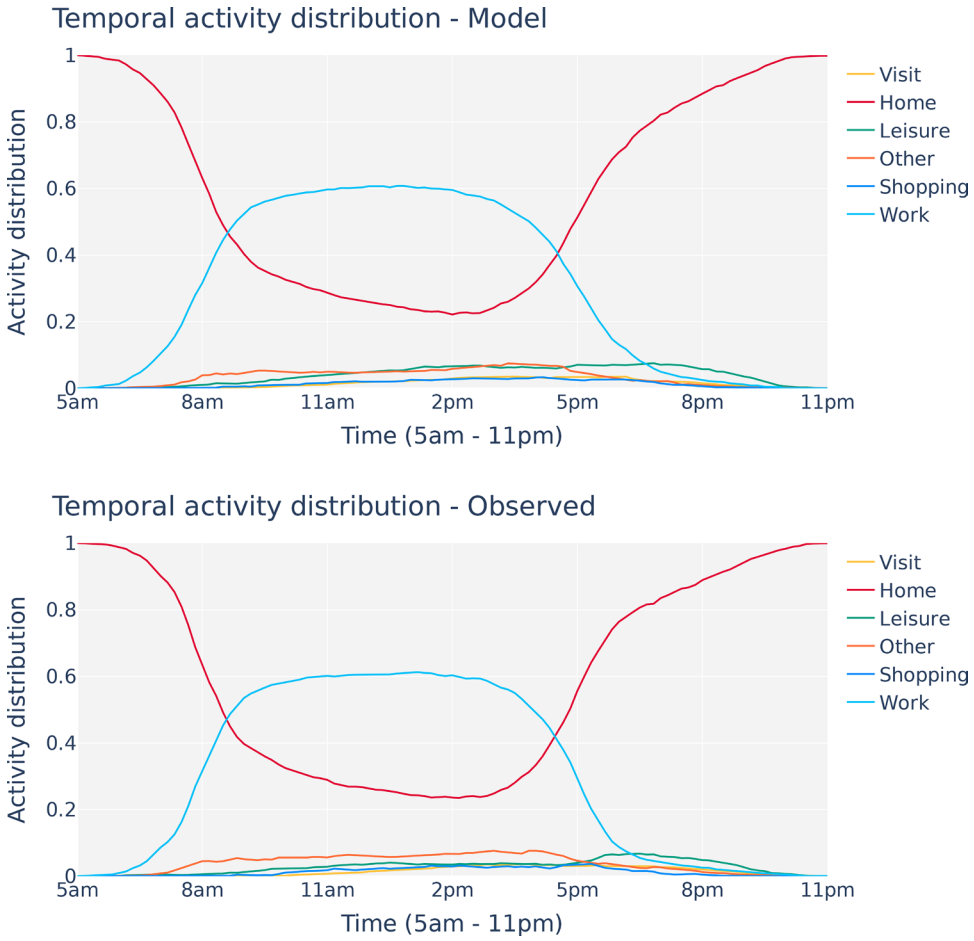


Fig. 7 Distribution of ongoing activities, performed between 5:00 am and 11:00 pm for simulated agents and observed individuals

agreement between predicted and observed outcomes. This is expected given their frequency in the data and the inclusion of the *been to work* feature, which provides contextual information about whether work has already occurred. In contrast, free-time activities are more difficult to predict accurately, which is reasonable since few input features explicitly indicate when these are likely to occur beyond the time of day.

For Mode Choice, both *public transport* and *car* are predicted with high accuracy. This distinction is likely driven by car- and PT-specific features such as costs and access variables that guide the model in the correct direction. Conversely, *walk* and *bike* decisions are less distinct and more difficult to separate, as they mainly rely on the absence of car ownership or a PT pass.

Ablation Experiment on Mandatory Travel Time and Cost

To assess the contribution of the Trip Distribution sub-model's forward-looking mandatory travel time and car cost sections described in Sects. 3.5.1 and 3.5.2, an ablation experiment was conducted. The sub-model was retrained without these features while keeping all other input features the same. Specifically, the features removed were: *travel time (origin mandatory)*, *travel time (zone mandatory)*, *car cost (origin mandatory)*, and *car cost (zone mandatory)*. A separate hyperparameter search was also preformed to ensure that the simplified model was optimized. The best-performing settings and training results for both model versions are presented in Table 5.

The test set performance for both versions is shown in Table 6. Removing the forward-looking features led to a test loss increase from 5.042 to 5.366 and a decrease in accuracy from 0.123 to 0.097. These results confirm that the inclusion of mandatory travel time and cost features meaningfully improves the model's ability to predict destination choices.

Sequential Error Propagation Results

This analysis builds on the simulation framework described in Sect. 5.2, using the simulated mode choice distribution as a basis for comparison. Table 7 compares the aggregate mode-share outcomes from the full-chain simulation and the OBS-D configuration, where observed destinations

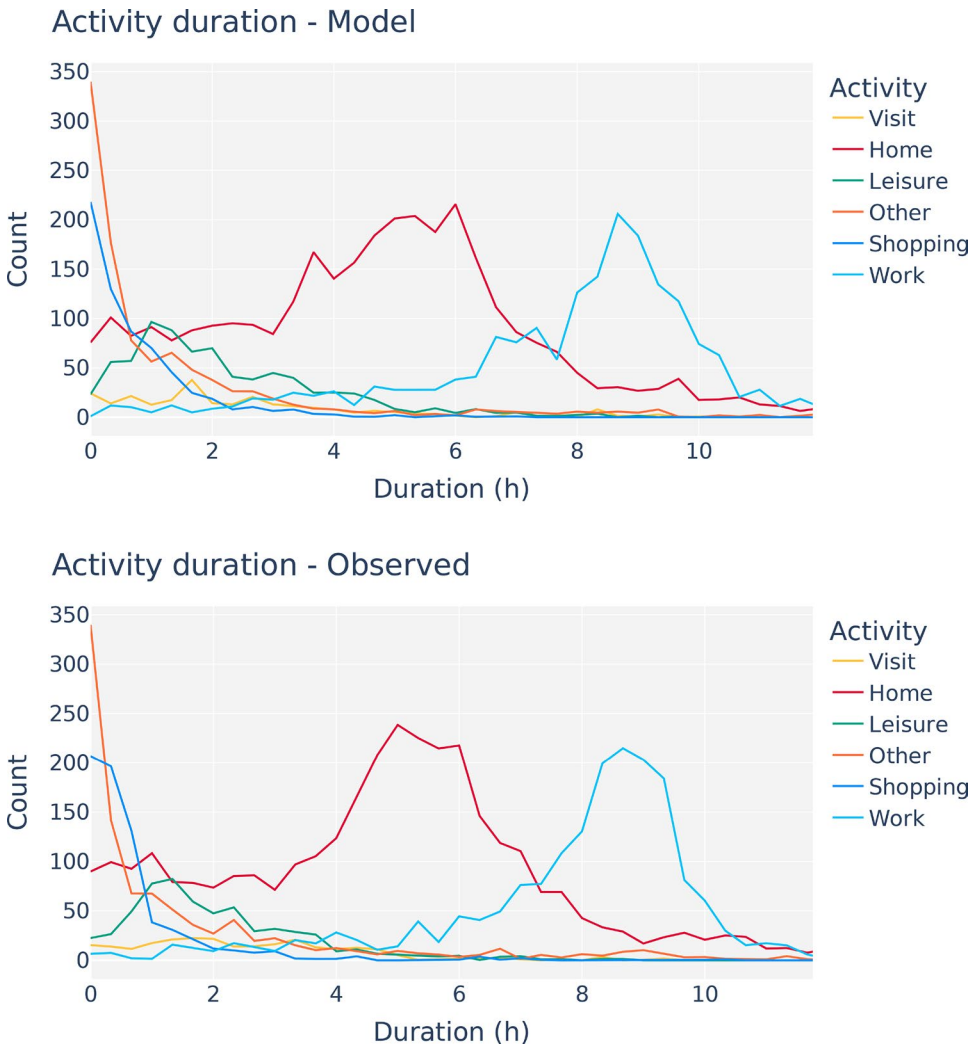


Fig. 8 Activity duration distribution for simulated agents and observed individuals. The data has been aggregated into bins to simplify visualization, with each bin representing a two consecutive time step interval. Note that the duration of the *home* activity is considered to end by the end of the day at 11:00 pm

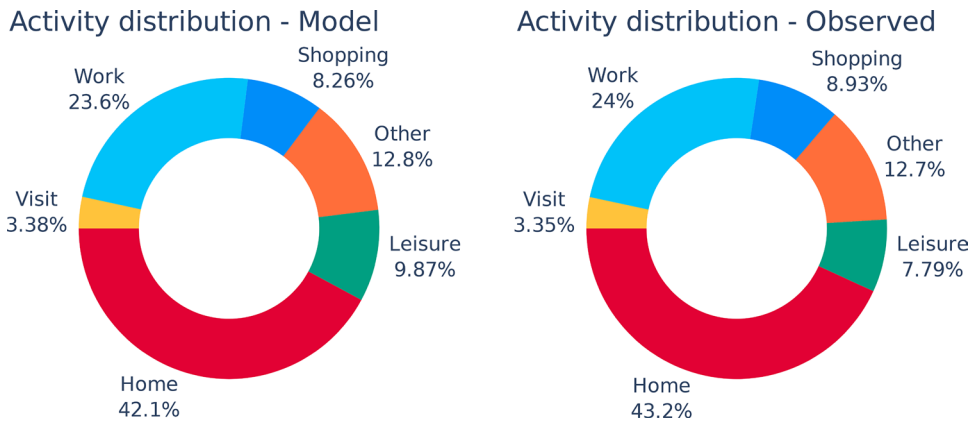


Fig. 9 Activity distribution for all trips among the simulated agents and observed individuals

were used as inputs to the Mode Choice model. Using observed destinations reduced the total variation distance from 0.0085 to 0.0050 (a 41.2% reduction), indicating that upstream prediction errors account for about two-fifths of the deviation between predicted and observed mode shares. Yet, since the absolute discrepancies are already small, the overall impact of error propagation on aggregate outcomes remains limited.

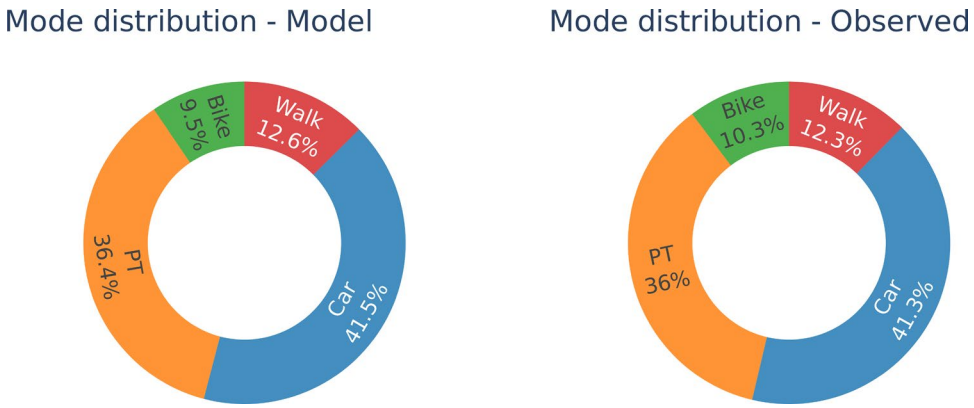


Fig. 10 Mode distribution for all trips among the simulated agents and observed individuals

Table 8 Trips and travel time—daily average

	Trips	Travel time
Model	2.74	66.3
Observed	2.67	66.7

Simulation Results

From the 10,819 individuals in the data, 25% were removed from the training and validation datasets and used entirely for testing the final model system. When trajectories had been simulated for all agent IDs in the test dataset, several statistics compared their simulated and corresponding

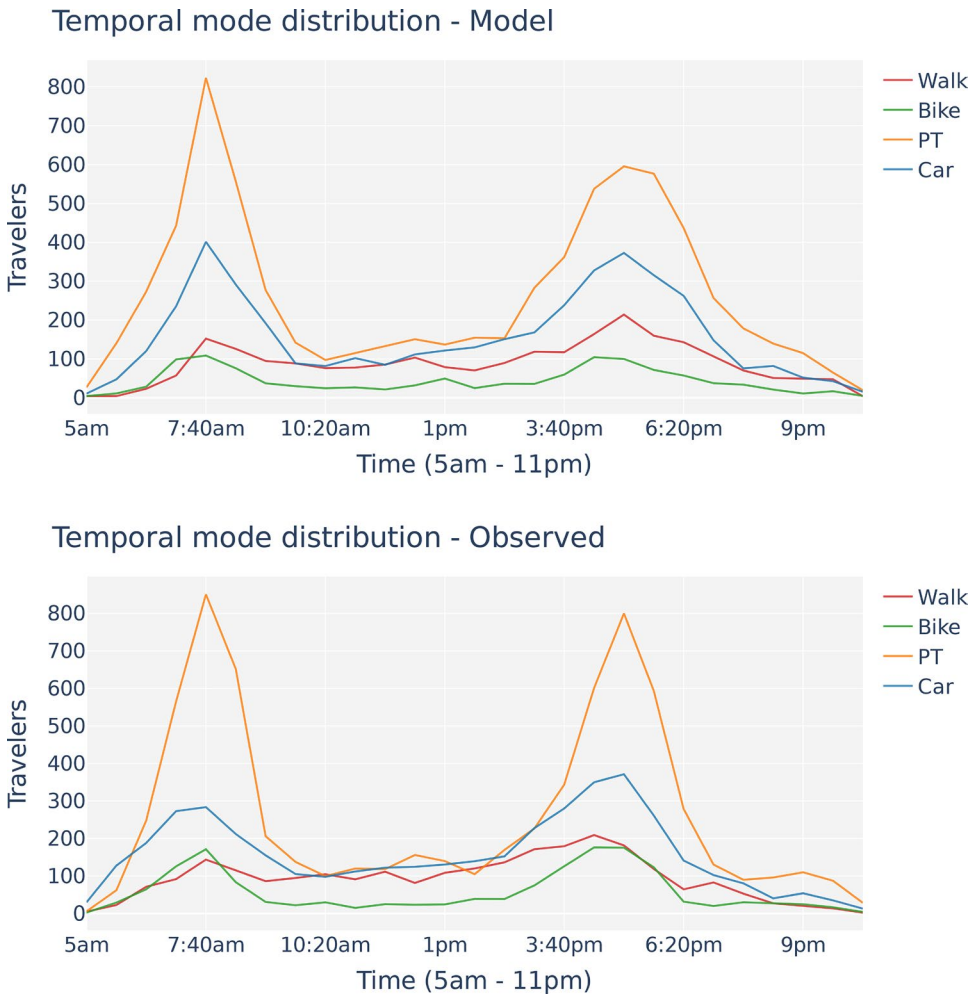


Fig. 11 The distribution of modes in use between 5:00 am and 11:00 pm for simulated agents and observed individuals. The data has been aggregated into bins to simplify visualization, with each bin covering four consecutive time steps.

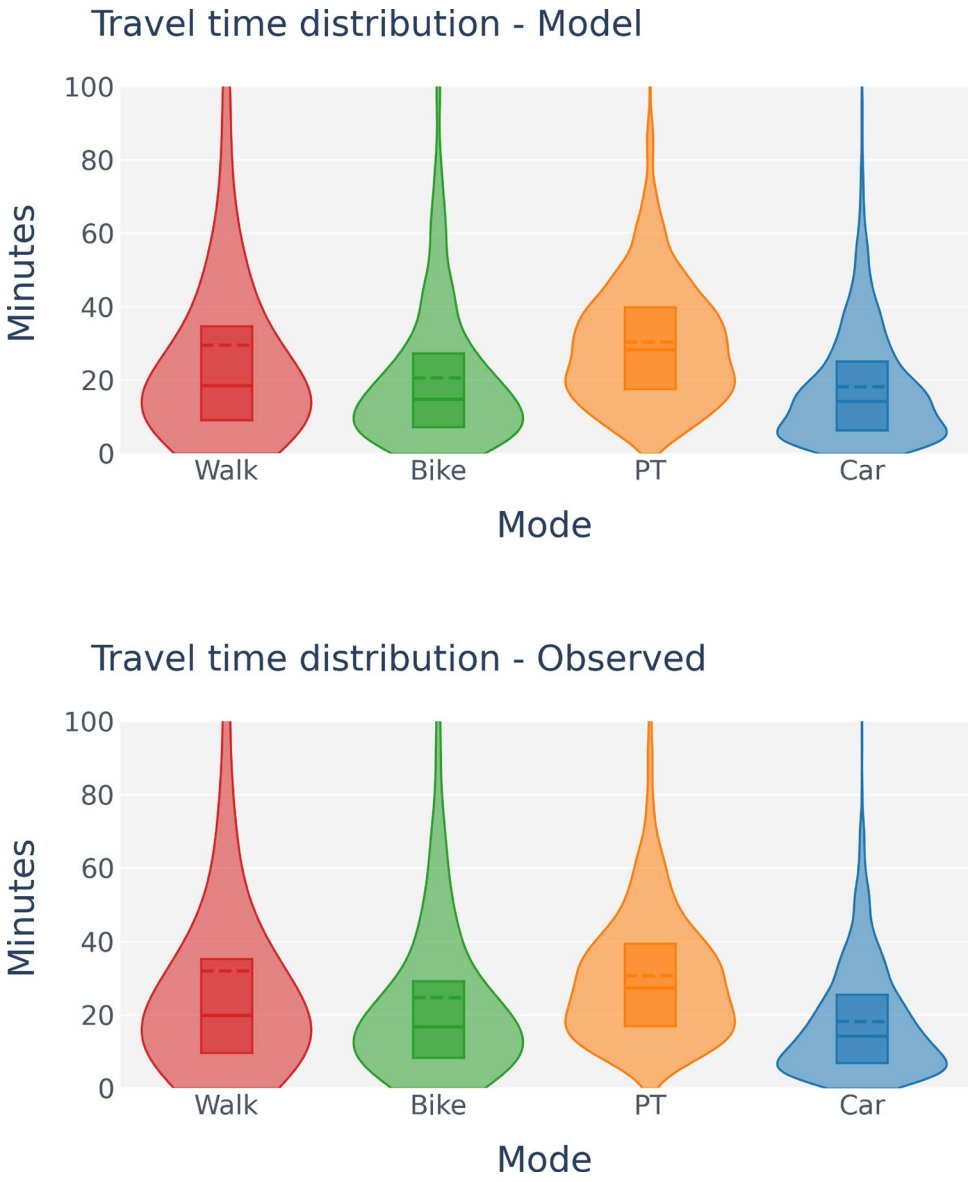


Fig. 12 Distributions of trip length in minutes for every mode for both model simulated agents and observed individuals. The mean line is dotted and the median line is filled

individual ID observed/ground truth daily schedules. All results are based on weighted individuals and agents to ensure representativeness.

Noteworthy is that the results from the temporal activity distribution, the activity duration distribution, and the activity distribution presented in Figs. 7, 8, and 9, respectively, are primarily influenced by the Trip Generation sub-model.

While the destination and transportation mode decisions are sampled from subsequent sub-models, they have minimal impact on the Trip Generation sub-model's activity-generated trips. Since the Trip Generation sub-model samples the start of an activity-based trip, its distribution of trip purposes inherently assumes that the agent will arrive within a reasonable time, ensuring that the activity remains feasible to do in the time slot upon arrival. The subsequent sub-models could impact activity start times if travel times are greatly misrepresented, causing activities to begin later than observed and leaving less time in the day for additional activities. Even with a perfect Trip Generation sub-model, this could lead to a mismatch in the simulated and observed activity distributions throughout the day. However, as long as the trip durations are not too different, this would have minimal implications on the aforementioned results.

All Trip Generation-based results align well with observed trajectories. The temporal activity distribution, shown in Fig. 7, demonstrates that the sub-model captures a well-represented distribution of activities throughout the day.

The activity duration distribution in Fig. 8 shows how the Trip Generation sub-model manages quite accurately to

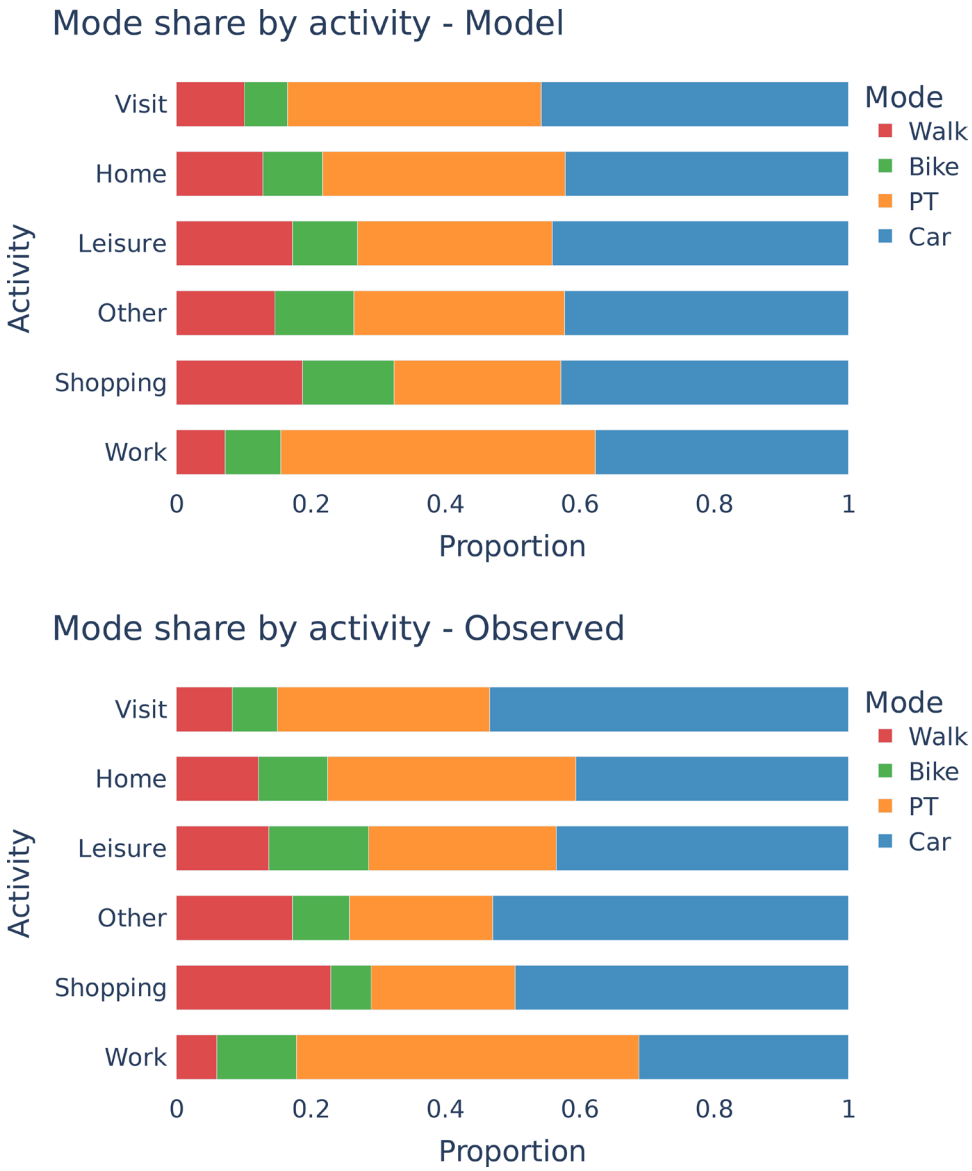


Fig. 13 Mode share by activity among the simulated agents and observed individuals

Table 9 Best hyperparameter results—Helsinki. Learning rate (LR), batch size (BS), hidden layers (H1-H2), validation loss (VL), training loss (TL), validation accuracy (VA), training accuracy (TA), epoch (E), and total training time in seconds (Sec)

Sub-model	LR	BS	H1	H2	VL	TL	VA	TA	E	Sec
Trip Generation	0.01	512	32	32	0.160	0.158	0.965	0.966	113	484
Trip Distribution	0.01	16	16	16	4.277	4.364	0.202	0.199	90	307
Mode Choice	0.01	64	16	16	0.659	0.651	0.778	0.776	106	114

learn these distributions as well. This can largely be attributed to its *time of day* and *duration* features, where the latter tracks the progress of the activity's duration and assists the sub-model in predicting the *stay* alternative accordingly.

The third result, primarily based on the Trip Generation sub-model, is the activity distribution among all trip purposes in Fig. 9. Some smaller discrepancies can be found in the comparisons.

To give a general sense of mode share over the day, Fig. 10 shows the mode distribution for all trips. Most significant is how *bike* is just slightly under-represented compared to the observed usage.

Table 8 presents the daily number of trips and time traveled, on average for each agent.

The distribution of transportation modes used throughout the day is displayed in Figure 11. It can be noted how simulated *PT* usage is a bit smaller than observed usage during afternoon peak hours.

Figure 12 shows the distributions of the trip duration in minutes for every mode, where mean and median lines seem to agree fairly well.

Figure 13 shows the mode share by activity. It can be seen that the distributions vary in multiple categories, which is expected since activity and mode choice are never modeled jointly. Activities are first selected in the Trip Generation sub-model, after which mode share is decided in the Mode Choice sub-model, without explicitly including the selected activity as an input feature. If it had, it could very well have improved this result. However, the distributions are not too far off, and it can be noted how the Mode Choice sub-model has picked up the general pattern from its other input features, such as travel times to the selected destination, which implicitly hint at what activity has been selected.

Implementing the Model to New Data—Helsinki

After training and evaluating the model on the Stockholm dataset, this section uses a new survey dataset collected in Helsinki in 2018, to briefly showcase its applicability, and transferability, here referring to the ability to adapt and retrain the architecture on data from another city.

Data

The Helsinki dataset featured 7,352 individuals and days and used a larger number of zones, 1,977, compared to Stockholm's 1375. The dataset also included Helsinki-specific data, such as land-use information, cost and travel time matrices. The transportation modes *car* and *PT* had, just like in the Stockholm scenario, additional travel time matrices for peak hours, applied during 6:30 am–8:30 am and 3:30 pm–5:30 pm. The land-use data used in the Trip Distribution sub-model was reduced from five to four categories: *total population*, *total employment*, *employment retail*, and *employment leisure*.

Monetary Cost

Since Helsinki did not apply a toll system for cars, the car cost function was reduced to only be based on distance. However, while Stockholm employs a constant public transport fare, Helsinki applies a zone-based system. Consequently, this variability allowed public transport costs to be incorporated into the Helsinki model. Similar to how car cost was implemented into the Stockholm model, the public transport cost added three new input features, *PT cost (origin zone)*, *PT cost (origin mandatory)*, and *PT cost (zone mandatory)* in the Trip Distribution sub-model and *PT cost to selected zone* in the Mode Choice sub-model.

Additional Features and Labels

In addition to the activities included in the Stockholm Trip Generation sub-model, a seventh activity labeled *school* was added in the Helsinki scenario. As a result, its Trip Generation sub-model included an additional output node for *school*, corresponding to each school-based trip. Moreover, the *occupation* feature section in the Trip Generation sub-model was expanded to include an additional input variable, now allowing for the identification of agents as workers, students, or neither.

In the Helsinki scenario, *school* is modeled as a separate mandatory activity for students, distinct from *work*, which remains mandatory only for workers. Similar to how the *work* activity was considered mandatory for workers, the *school* activity was mandatory for students. The feature *been to work* in the Trip Generation sub-model was redefined as *been to mand*, indicating whether a worker or a student had participated in their mandatory activity (*work* or *school*). If a student had not been to school yet, the Trip Distribution sub-model would calculate travel time and cost section features using the school zone as the mandatory destination and default to the home zone otherwise.

Lastly, unlike the Stockholm scenario, which calculated *walk* and *bike* travel times as a function of distance, Helsinki employed specific travel time matrices for both modes. Consequently, *walk* and *bike* did not have identical normalized features and used their unique travel times, leading to four specific travel time input variables in each travel time section.

Training and Evaluation

The model underwent the same training procedure as the Stockholm model, as outlined in Sect. 4. Different combinations of hyperparameters were explored. The best-performing settings were used to retrain each sub-model. These settings and their respective training results are shown in Table 9.

The training history, showing validation loss and accuracy progression per epoch for each retrained sub-model using the best-performing settings, is presented in Fig. 16 in Appendix C. Dimensions of all parameter matrices and vectors, including hidden, input, and output vectors, are shown in Table 11 in Appendix E.

Helsinki Model Results

The results from the Helsinki simulated activity-travel paths are shown in Appendix D. Similar to the Stockholm model, the Helsinki model accurately models the temporal activity distribution (Fig. 17), a result primarily based on the Trip Generation sub-model. The activity duration distribution in Fig. 18 has captured most of the observed distributions. The model underrepresents some of the shorter durations of the *Other* activity. The last result, primarily based on the Trip Generation sub-model, the activity distribution, is shown in Fig. 19. As the first sub-model in the sequence, the Trip Generation sub-model is well positioned to accurately simulate the true distribution of activities, as shown by the results.

The numbers of trips and travel time in minutes per day on average for each agent are shown in Table 10. The simulated mode distribution (Fig. 20) shows some over-representation for *PT* and underrepresentation for *car*.

The temporal mode distribution in Fig. 21 shows mostly similar behavior, except that observed *PT* trips are used less frequently than simulated trips from 3:40 pm onwards.

The travel time distribution in Fig. 22 shows close agreement between the simulated and observed distributions. However, the simulated median for walking is slightly lower than in the observed distribution.

The mode share by activity results is shown in Fig. 23. Similar to the Stockholm model, the Helsinki model can capture the general pattern of joint activity-mode distributions without using the selected activity as an input feature.

Discussion

The model system follows a sequential approach that separates the decision into three phases: Trip Generation, Trip Distribution, and Mode Choice. This approach has the advantage of separating model design into smaller parts, which can give more control and clearer insight into each sub-model. For example, the simulation results shown in Figs. 7, 8, and 9, covering the temporal activity distribution, activity duration distribution, and overall activity distribution, are primarily determined by the Trip Generation sub-model, providing clarity in how this part of the full model system performs. See Sect. 6 for more detail.

Another advantage is that this modeling structure is computationally efficient. Modeling everything jointly would require a more complex and larger model, using more parameters and taking longer time to train. Furthermore, if data is sparse, a more complex model could overfit as each alternative can connect with all input features, including those that may not be meaningful descriptors. A separation into sub-models reduces this risk by focusing on more specific relationships.

This separation of decision components, while beneficial in terms of structure and computational efficiency, comes with trade-offs. It fails to capture the joint nature of real-world decision-making, where individuals typically consider activity, destination, and transportation mode simultaneously. Furthermore, a sequential model system is also vulnerable to error propagation, where inaccuracies in early-stage decisions (e.g., Trip Generation) can cascade and distort the inputs to downstream sub-models, such as destination or mode choice.

The error propagation analysis showed that using observed destinations instead of predicted ones reduced total variation in mode shares by about 40%, indicating that upstream prediction errors contribute modestly to overall deviations. However, the absolute differences were already small, suggesting limited practical impact.

A strength of the model is its neural network design. The data-driven nonlinear transformations of the input variables play a significant role in enhancing the predictive performance of each sub-model. The attractiveness of an alternative can be based on complex combinations of multiple input features. For instance, the Trip Generation sub-model's decision of whether to stay can now be dependent on the combination of timing (from the feature *time of day*) and how long the current activity has prolonged

(*activity duration*). This, in turn, may have completely different outcomes depending on the agent's occupation (*occupation*). Likewise, the Trip Distribution sub-model is also able to find complex patterns, such that people may be more willing to travel long by car (*travel times*), for the leisure activity (*activity to be performed*), taking place in wooded or lakeside zones (*land use*). It becomes apparent that the numerous feature interdependencies included in scheduling a day greatly benefit from the nonlinear data-driven approach.

Another important design feature is the inclusion of the forward-looking travel time and cost features for each zone, allowing the model to evaluate how visiting a zone would impact future (mandatory) travel. An ablation experiment showed that removing these features from the Trip Distribution sub-model led to a noticeable increase in cross-entropy loss and a drop in accuracy, as shown in Sect. 6.3. This confirms that incorporating information about upcoming mandatory activities significantly enhances the model's destination evaluations.

Future work could explore a more integrated modeling approach in which activity, destination, and mode choice decisions are evaluated jointly rather than through separate sub-models. Another promising direction is the incorporation of components that explicitly model the sequential nature of daily travel behavior. For example, Long Short-Term Memory (LSTM) layers or attention-based mechanisms such as Transformers could be valuable for capturing temporal dependencies in the decision-making process. Recent studies have demonstrated the effectiveness of such architectures for modeling travel demand patterns, including short-term passenger demand prediction (Ghandeharioun et al. 2023), urban mobility (Mathivanan et al. 2024), and daily human mobility sequences (Wang and Osaragi 2024). In addition, graph-based neural network models may be used to capture spatial dependencies in destination choice, where zone attractiveness can be influenced by neighboring opportunities (Xue et al. 2025; Wang et al. 2024). Finally, future studies could incorporate formal sensitivity analyses (e.g., systematic variations in travel times, costs, or population characteristics) to assess behavioral robustness in policy and infrastructure scenarios.

Considering that the proposed sequential approach is both more computationally efficient and easier to interpret while being a beneficial framework for sparse datasets, Skyline-NN provides a solid foundation for a fully neural network-based scheduling and travel demand model that includes activity, destination, and mode choice alternatives.

Conclusion

This paper presented Skyline-NN, a fully neural network-based travel demand and scheduling model that sequentially simulates individuals' activity, destination, and transportation mode choices at each time step throughout the day. By dividing decisions across three sequentially employed sub-models, Trip Generation, Trip Distribution, and Mode Choice, Skyline-NN reduces the complexity and computational burden associated with fully joint modeling. This framework also lowers the risk of overfitting in sparse data settings and provides clear insight into the behavior of each sub-model. The model achieves good performance across multiple metrics when comparing simulated and observed daily schedules. It also outperformed corresponding multinomial logit baselines across all sub-models, underscoring the benefits of the neural network formulation in capturing nonlinear relationships.

However, the disjoint nature of the approach introduces limitations. It may overlook interactions that naturally occur when individuals consider multiple factors simultaneously, and the accuracy of subsequent sub-models may be influenced by earlier ones. The error propagation analysis indicated that these effects were minor. Replacing predicted destinations with observed ones reduced total variation in mode shares by about 40%, yet the absolute deviations were small, suggesting limited practical significance.

Despite these challenges, the architecture's design, featuring a fully neural network-based structure with data-driven nonlinear transformations and shared zone parameters, contributes to a model with good performance. Its forward-looking mandatory travel time and cost features further improve model accuracy, as confirmed by an ablation experiment. Skyline-NN provides a solid, fully neural network-based foundation for modeling daily travel behavior and offers a valuable tool for policy analysis.

Appendix A: Map Visualization of Choice Probabilities

See Fig. 14,

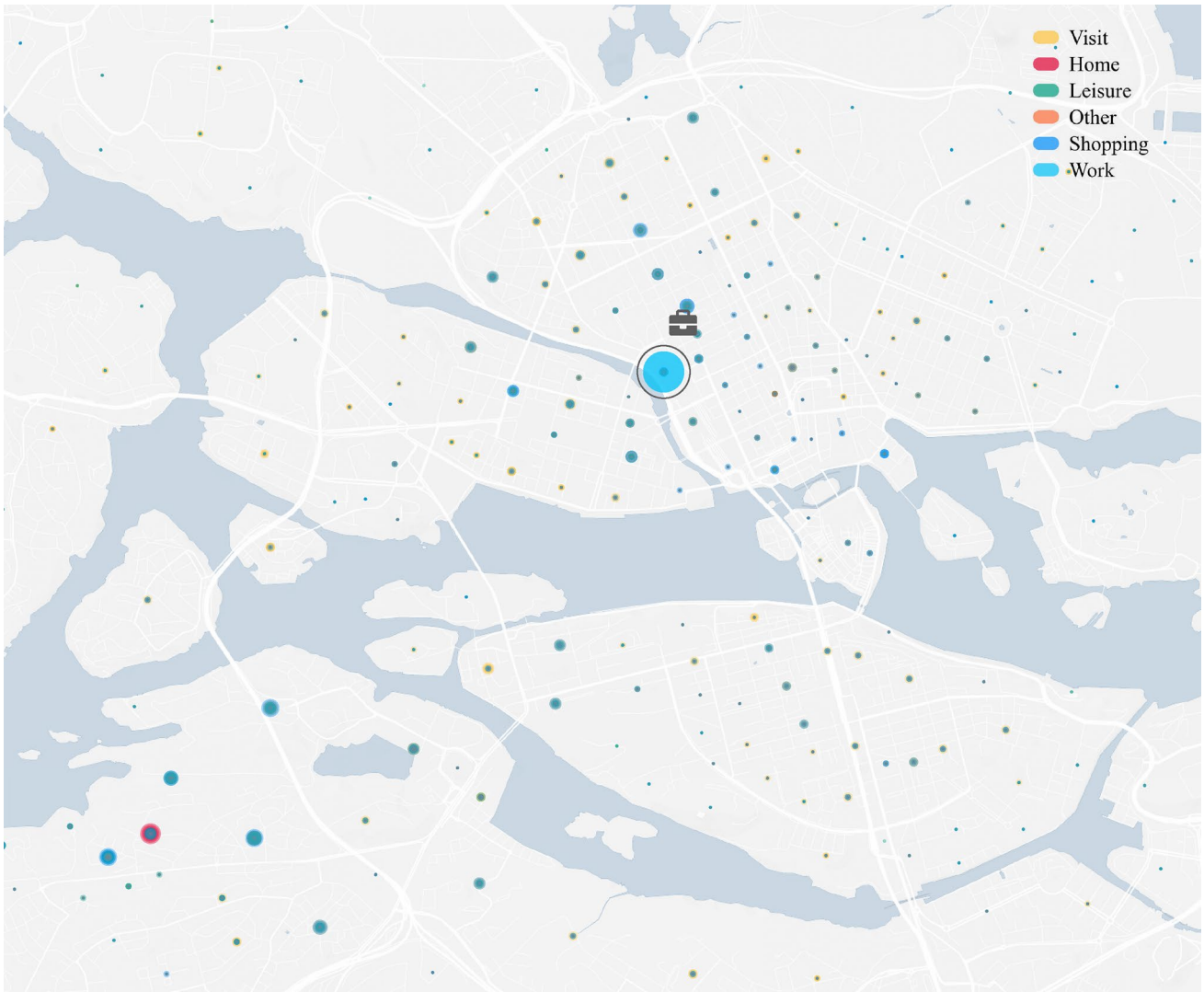


Fig. 14 The agent is currently in central Stockholm, engaged in the *work* activity (indicated by the black circle and briefcase icon). The colored circles show the probability of selecting any activity-destination pair, aggregated across all transportation modes and time steps between 4:20 pm and 4:50 pm, with circle size representing relative probability. Probabilities for *home*, *work*, and same zone (*stay*) have been scaled down. Notably, distant zones around the home (red circle) show high probabilities, influenced by mandatory travel time sections that favor zones closer to the next mandatory destination, here being the home zone

Appendix B: Training History—Stockholm

See Fig. 15.



Fig. 15 Training history of all Stockholm sub-models, showing the loss and accuracy for each epoch for both training and validation data

Appendix C: Training History—Helsinki

See Fig. 16.

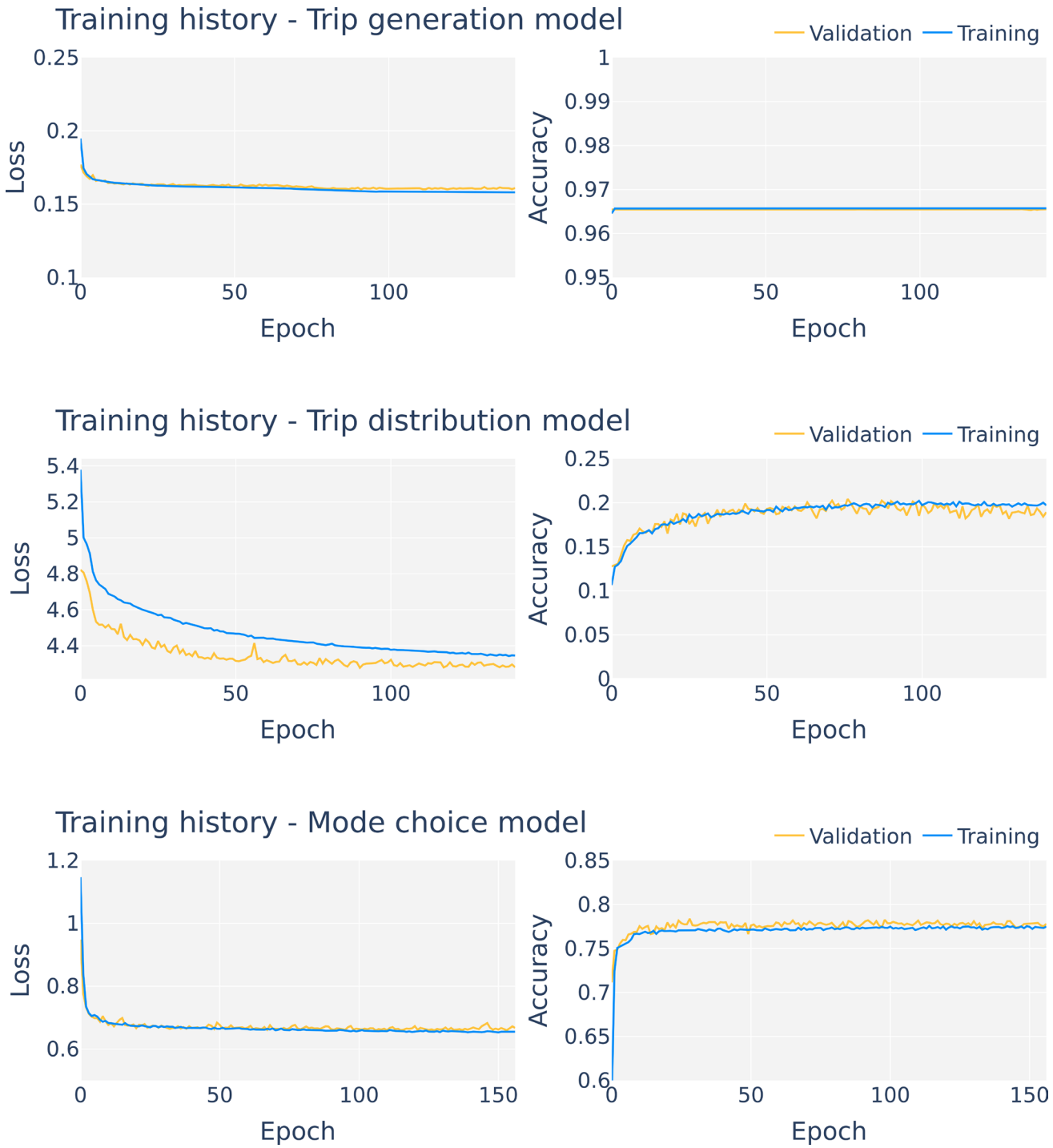


Fig. 16 Training history of all Helsinki sub-models, showing the loss and accuracy for each epoch for both training and validation data

Appendix D: Helsinki Results

See Fig. 17, 18, 19, 20, 21, 22, 23 and Table 10.

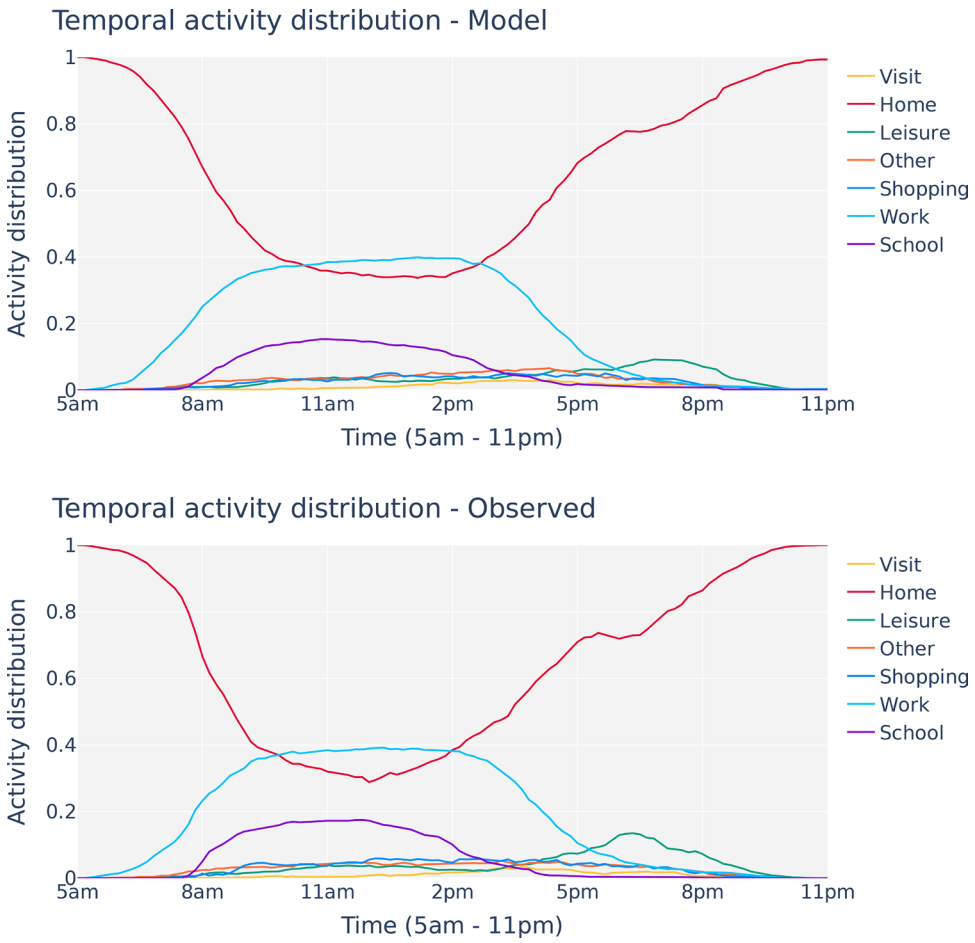


Fig. 17 Distribution of ongoing activities, performed between 5:00 am and 11:00 pm for simulated agents and observed individuals

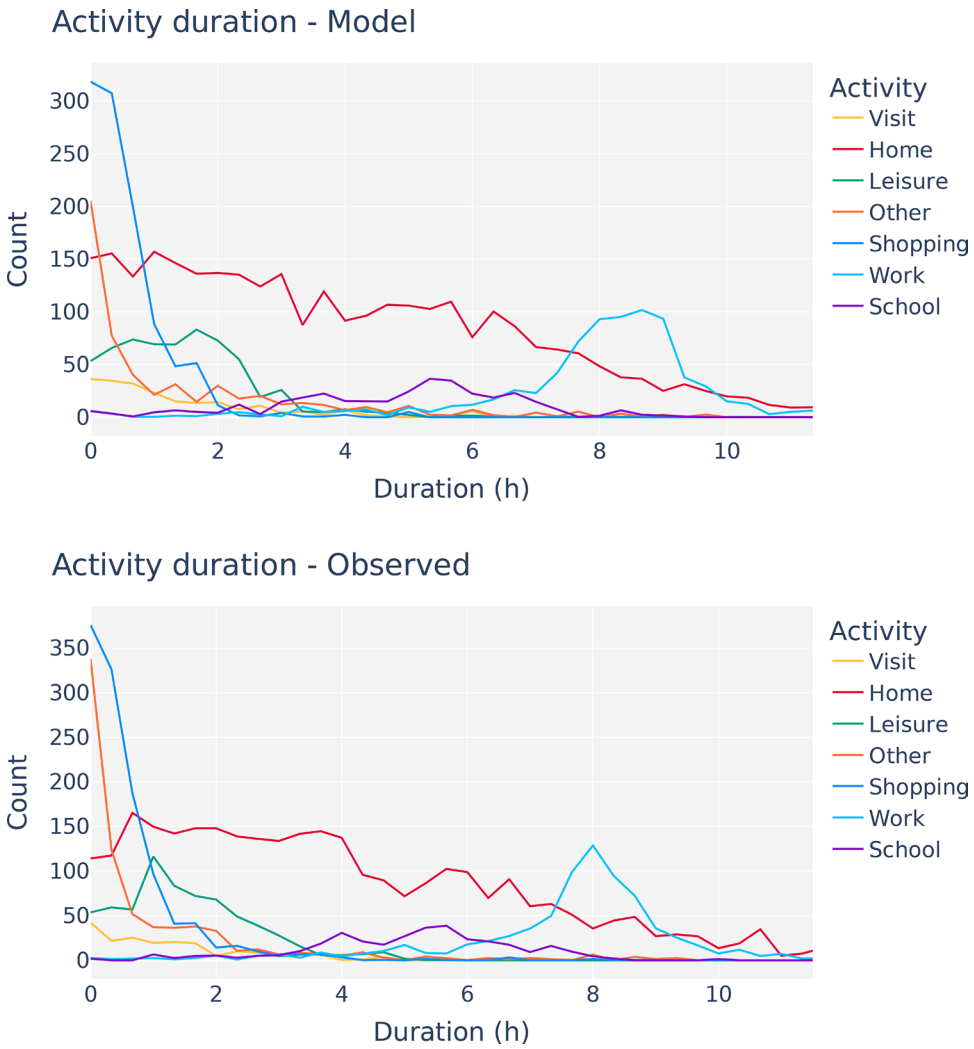


Fig. 18 Activity duration distribution for simulated agents and observed individuals. The data has been aggregated into bins to simplify visualization, with each bin representing a two consecutive time step interval. Note that the duration of the *home* activity is considered to end by the end of the day at 11:00 pm

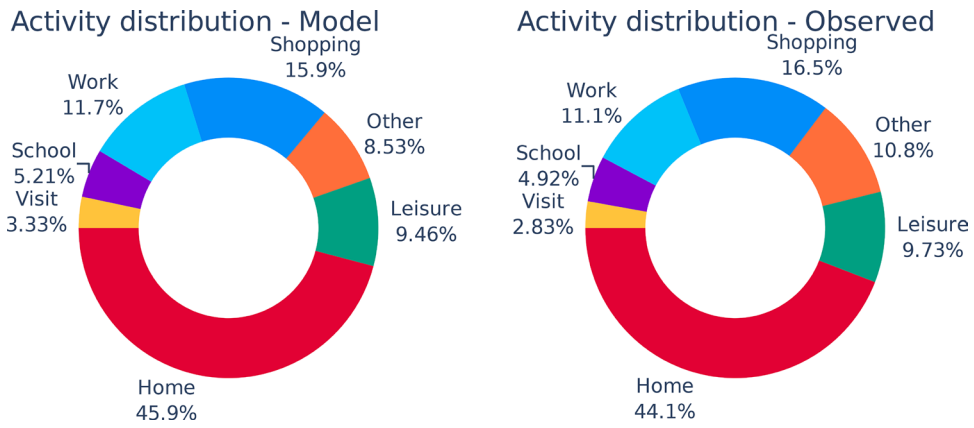


Fig. 19 Activity distribution for all trips among the simulated agents and observed individuals

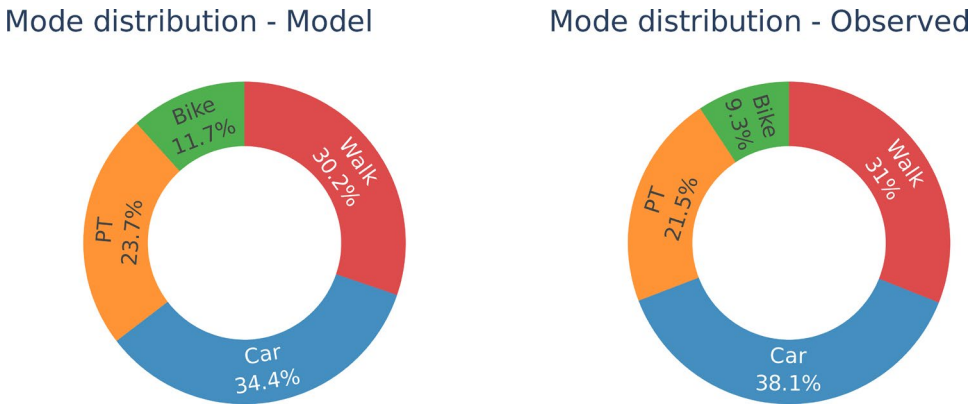


Fig. 20 Mode distribution for all trips among the simulated agents and observed individuals

Table 10 Trips and travel time—daily average

	Trips	Travel time
Model	3.56	49.7
Observed	3.72	50.8

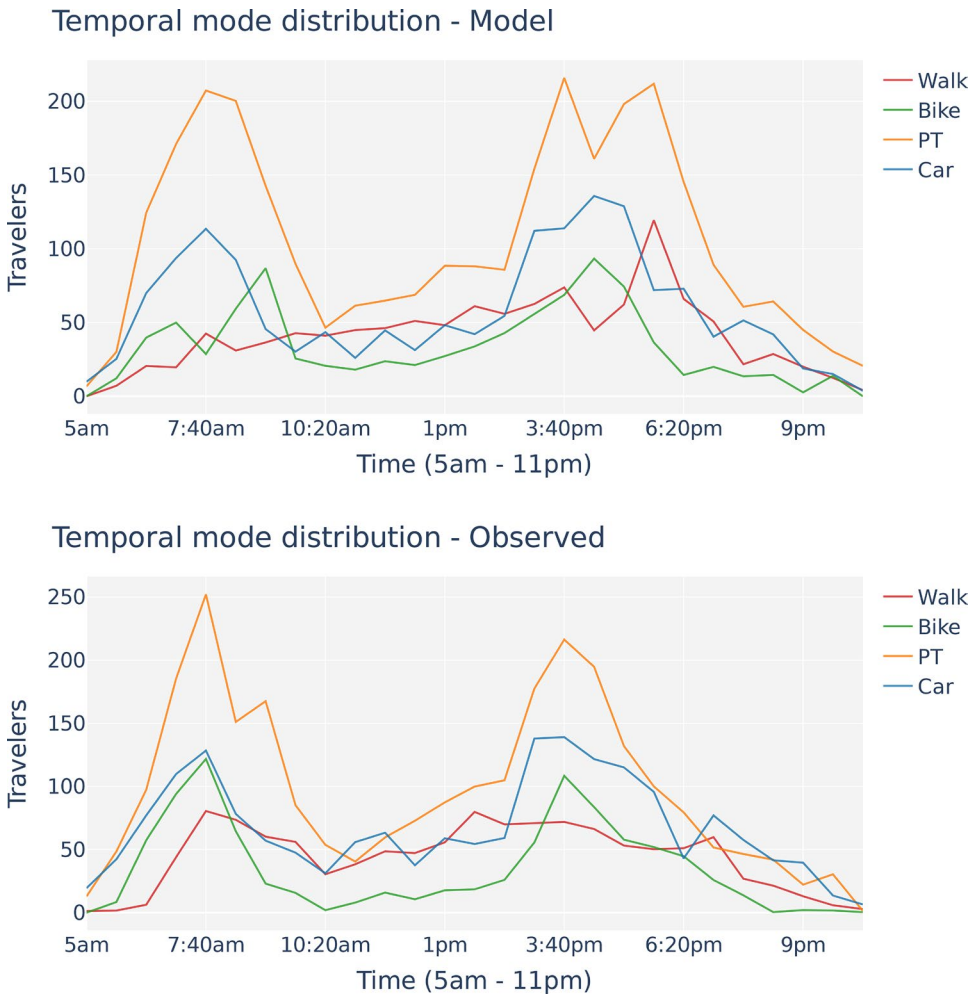


Fig. 21 Distribution of modes in use between 5:00 am and 11:00 pm for simulated agents and observed individuals. The data has been aggregated into bins to simplify visualization, with each bin covering four consecutive time steps.

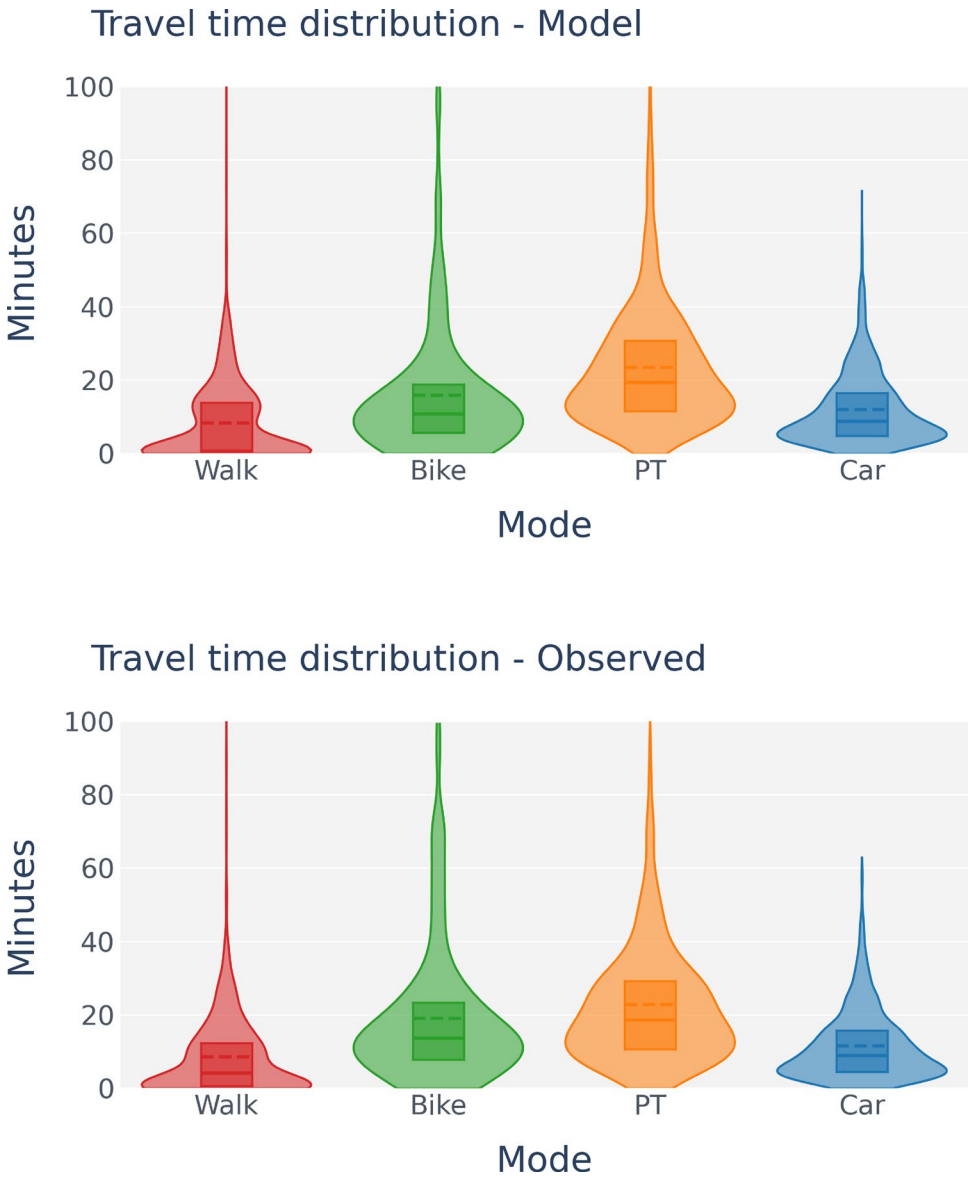


Fig. 22 Distributions of trip length in minutes for every mode for both model simulated agents and observed individuals. The mean line is dotted and the median line is filled

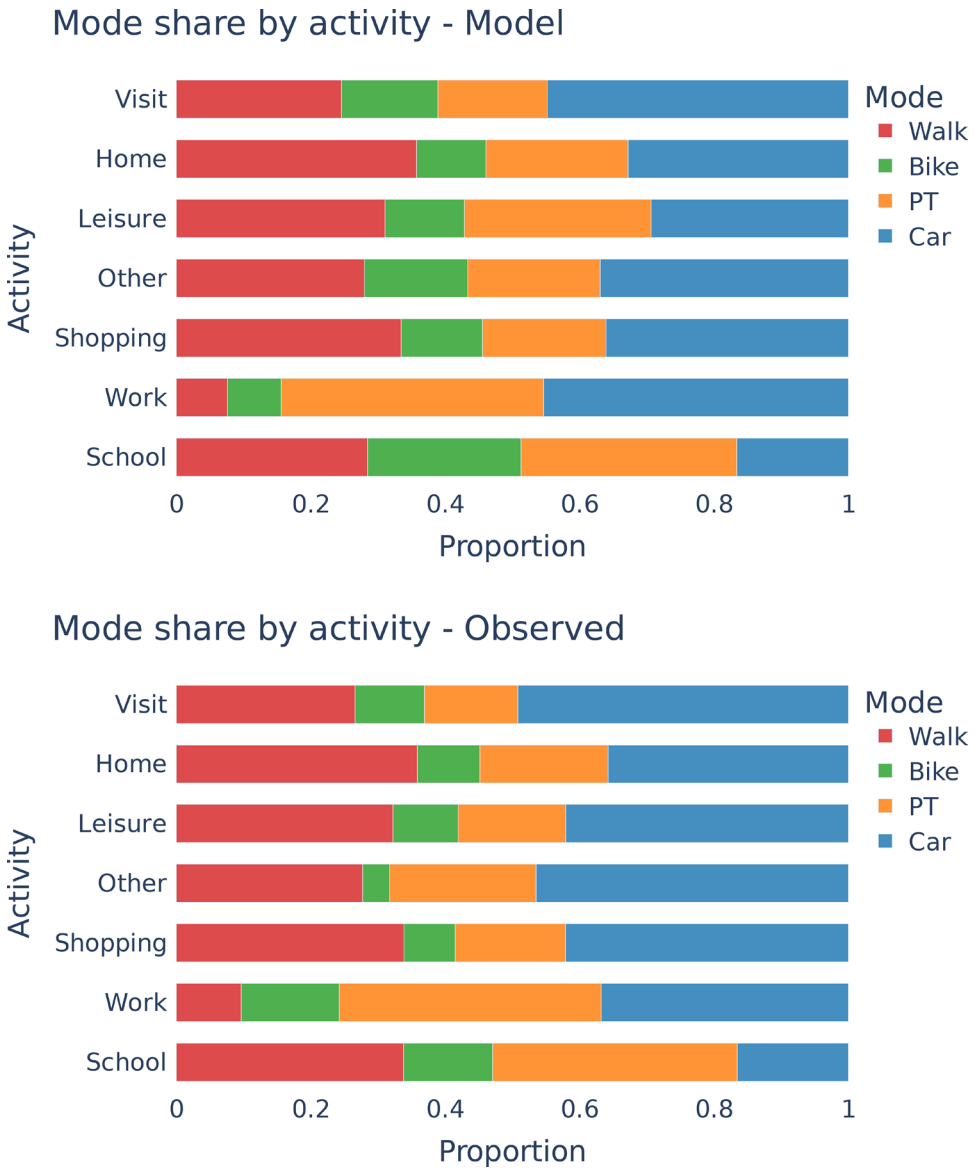


Fig. 23 Mode share by activity among the simulated agents and observed individuals

Appendix E: Helsinki Parameters and Vector Dimensions

See Table 11,

Table 11 Dimensions of parameters and vectors for all sub-models—Helsinki. The specifications for the Trip Distribution sub-model refer to the operations in the Zone Block Module, with an input vector representing the features for the zone that the zone block is evaluating

Sub-model	Input	Hidden layer 1			Hidden layer 2		
	x	W	b	h	W	b	h
Trip generation	12×1	32×12	32×1	32×1	32×32	32×1	32×1
Trip distribution	29×1	16×29	16×1	16×1	16×16	16×1	16×1
Mode choice	9×1	16×9	16×1	16×1	16×16	16×1	16×1

	Output layer		
	W	b	$\hat{y}$
Trip generation	8×32	8×1	8×1
Trip distribution	1×16	1×1	1×1
Mode choice	4×16	4×1	4×1

Author Contributions J.F. was responsible for conceptualization, methodology, software, formal analysis, investigation, validation, data curation, visualization, writing—original draft, and writing—review and editing. A.K. contributed to supervision, resources, data curation, and writing—review and editing.

Funding Open access funding provided by Royal Institute of Technology. No funding was received for conducting this study.

Data Availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of Interest The authors declare no conflict of interest.

Ethical Approval Not applicable.

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References

Aboutaleb YM, Danaf M, Xie Y, Ben-Akiva M (2021) Discrete choice analysis with machine learning capabilities. ArXiv Preprint. https://doi.org/10.48550/arXiv.2101.10261

Alwosheel A, Cranenburgh S, Chorus CG (2021) Why did you predict that? towards explainable artificial neural networks for travel demand analysis. Transp Res Part C 128:103143. https://doi.org/10.1016/j.trc.2021.103143

Arentze T, Hofman F, Mourik H, Timmermans H (2000) Albatross: multiagent, rule-based model of activity pattern decisions. Transp Res Rec 1706(1):136–144. https://doi.org/10.3141/1706-16

Bei H, Chen H, Li L, Gao X, Xia Y, Sun Y (2023) Joint prediction of travel mode choice and purpose from travel surveys: a multitask deep learning approach. Travel behaviour and society 33:100625. https://doi.org/10.1016/j.tbs.2023.100625

Bentz Y, Merunka D (2000) Neural networks and the multinomial logit for brand choice modelling: a hybrid approach. J Forecast 19(3):177–200

Bowman JL, Ben-Akiva ME (2001) Activity-based disaggregate travel demand model system with activity schedules. Transp Res Part A 35(1):1–28. https://doi.org/10.1016/S0965-8564(99)00043-9

Castiglione J, Bradley M, Gliebe J (2015) Activity-based travel demand models: a primer vol SHRP 2 report S2-C46-RR-1. The National Academies Press, Washington, DC. https://doi.org/10.17226/22357

Chapin FS (1974) Human activity patterns in the city: things people do in time and in space. Wiley, New York

Chu Z, Cheng L, Chen H (2012) A review of activity-based travel demand modeling. CICTP, 2012: multimodal transportation systems-convenient, safe, cost-effective, efficient. CICTP, pp 48–59. https://doi.org/10.1061/9780784412442.006

Cranenburgh S, Wang S, Vij A, Pereira F, Walker J (2022) Choice modelling in the age of machine learning-discussion paper. J Choice Modell 42:100340. https://doi.org/10.1016/j.jocm.2021.100340

Drchal J, Čertický M, Jakob M (2019) Data-driven activity scheduler for agent-based mobility models. Transp Res Part C 98:370–390. https://doi.org/10.1016/j.trc.2018.12.002

Fredriksson J, Karlström A (2023) Analyzing non-linear contributions to predictive performance in a neural network based scheduling model. Proced Comput Sci 220:680–685. https://doi.org/10.1016/j.procs.2023.03.088

Ghandeharioun Z, Zendehdel Nobari P, Wu W (2023) Exploring deep learning approaches for short-term passenger demand prediction. Data Sci Transpo 5(3):19. https://doi.org/10.1007/s42421-023-00083-w

Glorot X, Bengio Y (2010) Understanding the difficulty of training deep feedforward neural networks. In: Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics, pp. 249–256

Goodfellow I, Bengio Y, Courville A (2016) Deep learning. MIT Press, Cambridge

Hagenauer J, Helbich M (2017) A comparative study of machine learning classifiers for modeling travel mode choice. Expert Syst Appl 78:273–282. https://doi.org/10.1016/j.eswa.2017.01.057

Hägerstrand T (1970) What about people in regional science. Reg Sci Assoc. https://doi.org/10.1111/j.1435-5597.1970.tb01464.x
James G (2013) An introduction to statistical learning. Springer, New York
Karlström A (2005) A dynamic programming approach for the activity generation and scheduling problem. Springer, Cham, pp 25–42. https://doi.org/10.1016/B978-008044581-6/50005-3
Kingma DP, Ba J (2014) Adam: a method for stochastic optimization. ArXiv Preprint. https://doi.org/10.48550/arXiv.1412.6980
Kitamura R, Kermanshah M (1984) Sequential model of interdependent activity and destination choices. Transp Res Rec 987:81–89
Koushik A, Manoj M, Nezamuddin N, Prathosh AP (2023) Activity schedule modeling using machine learning. Transp Res Rec 2677(8):1–23. https://doi.org/10.1177/03611981231155426
LeCun Y, Bengio Y, Hinton G (2015) Deep learning. Nature 521(7553):436–444. https://doi.org/10.1038/nature14539
Mathivanan SK, Rajadurai H, Cho J, Easwaramoorthy SV (2024) A multi-modal geospatial-temporal lstm based deep learning framework for predictive modeling of urban mobility patterns. Sci Rep 14(1):31579. https://doi.org/10.1038/s41598-024-74237-3
McFadden D (1974) Conditional logit analysis of qualitative choice behavior. In: Zarembka P (ed) Frontiers in econometrics. Academic Press, New York, pp 105–142
McFadden D (1978) Modelling the choice of residential location. In: Karlqvist A (ed) Spatial interaction theory and planning models. Elsevier, Amsterdam
McNally MG (2007) The four-step model. Handbook of transport modelling. Emerald Group Publishing Limited, Bingley
Molnar C, Freiesleben T, König G, Casalicchio G, Wright MN, Bischl B (2021) Relating the partial dependence plot and permutation feature importance to the data generating process. ArXiv Preprint. https://doi.org/10.48550/arXiv.2109.01433
Omrani H (2015) Predicting travel mode of individuals by machine learning. Transp Res Proced 10:840–849. https://doi.org/10.1016/j.trpro.2015.09.037
Pineda-Jaramillo JD (2019) A review of machine learning (ml) algorithms used for modeling travel mode choice. Dyna 86(211):32–41. https://doi.org/10.15446/dyna.v86n211.79743
Rasouli M, Nikraz H (2013) Trip distribution modelling using neural network. In: Australasian Transport Research Forum, ATRF 2013 Proceedings
Salas P, Fuente R, Astroza S, Carrasco JA (2022) A systematic comparative evaluation of machine learning classifiers and discrete choice models for travel mode choice in the presence of response heterogeneity. Expert systems with applications 193:116253. https://doi.org/10.1016/j.eswa.2021.116253
Sifringer B, Lurkin V, Alahi A (2018) Enhancing discrete choice models with neural networks. In: Proceedings of the 18th Swiss Transport Research Conference (STRC), Monte Verità/Ascona, Switzerland, pp. 16–18
Timmermans HJP (2005) Progress in activity-based analysis. Elsevier, Amsterdam
Train KE (2009) Discrete choice methods with simulation. Cambridge University Press, Cambridge
Västberg O, Karlström A, Jonsson D, Sundberg M (2019) A dynamic discrete choice activity-based travel demand model. Transp Sci. https://doi.org/10.1287/trsc.2019.0898
Wang W, Osaragi T (2024) Learning daily human mobility with a transformer-based model. ISPRS Int J Geo Inf 13(2):35. https://doi.org/10.3390/ijgi13020035
Wang K, Peng Z, Cai M, Wu H, Liu L, Sun Z (2024) Coupling graph neural networks and travel mode choice for human mobility prediction. Physica A 646:129872
Xue J, Tan R, Ma J, Ukkusuri SV (2025) Data science in transportation networks with graph neural networks: a review and outlook. Data Sci Transp 7(2):1–27. https://doi.org/10.1007/s42421-025-00124-6
Zhang Z, Beck MW, Winkler DA, Huang B, Sibanda W, Goyal H (2018) Opening the black box of neural networks: methods for interpreting neural network models in clinical applications. Ann Transl Med. https://doi.org/10.21037/atm.2018.05.32
Zhang J, Hasan S, Roy KC, Yan X (2024) Predicting individual mobility behavior of ride-hailing service users considering heterogeneity of trip purposes. Data science for transportation 6(3):29. https://doi.org/10.1007/s42421-024-00113-1
Zhao X, Yan X, Yu A, Van Hentenryck P (2020) Prediction and behavioral analysis of travel mode choice: a comparison of machine learning and logit models. Travel Behav Soc 20:22–35. https://doi.org/10.1016/j.tbs.2020.02.003

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